SECTION VI

GENETICS OF KIDNEY DISEASE

44 Genetics of Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- A precise genetic diagnosis in a patient with Mendelian disease can provide diagnostic and prognostic information for the patient and allow predictive testing to be offered to family members.
- A genetic diagnosis can inform important clinical, therapeutic, life and reproductive decisions for a patient and family, sometimes facilitating transplant and reproductive interventions. This means that consequences of misinterpreting a genetic test result can be very serious indeed.
- Genetics and genomics professional organizations providing guidance for the pathogenicity classification of genetic variants stipulate the need for strong evidence for a variant to be clinically actionable.
- Understanding the prior likelihood of a type of genetic change being responsible for a patient's presentation is valuable in interpreting genetic test results.
- Discovery of the molecular basis of kidney diseases has deepened understanding of those conditions, informing how they are classified clinically, and has facilitated development of new treatments. As more treatments become available, the value of making a molecular diagnosis in individual patients is likely to increase.

INTRODUCTION: GENETICS, BIOLOGY, AND DISEASE

Genetic factors underpin much of biology because information needed for the formation, development, and a large part of the function of every organism is encoded by genetic material that is fixed from before embryogenesis and is present in almost all of an organism's cells throughout life. Technologic advances, especially the human genome project and massive parallel (or next generation) sequencing have resulted in an explosion in the understanding of how variation in this genetic information is associated with traits (i.e., phenotypes and diseases), and since most of this genetic information does not change during life, in most circumstances statistically robust genetic associations can be inferred to have a causal relationship with a trait. This has provided significant insights into the biology underlying numerous diseases, can inform the development of new treatments and even, in some cases, the use of therapies in individual patients.

However, new understanding from genetics research has emphasized the necessity for quantitative, rather than purely qualitative, understanding of how genetic variation affects disease risk. The quantum leap (brought about by the advent of massively parallel sequencing technologies) in our ability to ascertain genetic variation in individuals requires similar advancement in the understanding of what these newly accessible genetic variants mean for an individual if these new data are to be safely and effectively used in clinical practice.

MOLECULAR GENETICS IN MEDICINE

Since the last quarter of the 20th century, genetic testing has been increasingly used in clinical practice. In most situations, a genetic test is used to identify the precise genetic change responsible for a Mendelian disease in an individual or a family. Identification of such a "molecular diagnosis" can have numerous impacts on care or decision making for a patient or their wider family:

1. It can confirm the precise diagnosis, sometimes obviating the need for invasive investigations or a prolonged diagnostic odyssey and sometimes confirming the mode of transmission (including which relatives are potentially at risk);
2. It can provide clarity for a patient or family as to why they have developed a disease, and this can be psychologically helpful, although it can also lead to feelings of guilt among parents who may feel "responsible" for passing on a genetic change;
3. It can provide prognostic information, which in turn can help with life choices such as career or family planning (including a decision to adopt or not have children), can help decide whether treatment is indicated, and sometimes it can reveal the risk of disease recurrence following transplantation;
4. It can inform selection of specific therapy or obviate the need for therapies known to be ineffective (such as corticosteroid treatment for monogenic podocytopathy presenting with focal segmental glomerulosclerosis);
5. It can allow predictive testing to be offered to at-risk individuals. This, in turn, can:
   a. inform their key life decisions about career, financial, or family planning (whether or when to have children, or whether to use reproductive interventions);
   b. allow them to access health or life insurance that would otherwise be more expensive or unavailable (with a negative test in the setting of a positive family history, noting that in some jurisdictions (e.g., Australia and UK) a positive predictive test result is privileged information that need not be shared with insurers);
   c. allow access to (or discharge from) a surveillance program aimed at mitigating their risk from the disease;
   d. inform transplantation decisions, allowing or precluding safe donation of a kidney by an ostensibly unaffected relative;
   e. However, even where clinically or otherwise helpful or actionable, predicting a diagnosis in an otherwise healthy individual can have negative psychological consequences, sometimes characterized by patients with pervasive or unpleasant imagery of a "ticking time bomb,"[1] It is important to address this possibility before genetic testing where possible (e.g., see https://www.nhs.uk/conditions/predictive-genetic-tests-cancer/). This dilemma will be further highlighted and will need to be resolved in a scholarly and thoughtful manner as more groups are offered genome sequencing at a population level and genetic variants are revealed.
6. A predictive test can be used for reproductive interventions, including prenatal or preimplantation genetic diagnosis, allowing a couple naturally at high risk of having an affected child the opportunity to have an unaffected child.

The utility and impact (medically, psychologically, and socially) of a molecular diagnosis in an individual or family can therefore be enormous, and there has been much effort in delivering genetic testing to patients for this reason: There are numerous examples across the medical literature and popular press justifiably extolling the benefits of genetic testing. However, because of the extremely consequential decisions that patients, relatives, and clinicians can sometimes make on the basis of a genetic test result, it is **absolutely imperative** that the information yielded by such a test is accurate *and correctly interpreted by the clinician*. Erroneously actioning a genetic variant that confers only a low (or even zero) risk of a disease as if it is "the cause of disease" in an individual can have negative or even catastrophic consequences, both medical (such as unnecessary mastectomy in a young woman harboring only harmless variants in a *BRCA* gene, or mistakenly precluding donation of a kidney by a healthy parent to their child on dialysis owing to detection of a low- or no-risk genetic variant) or for the patient's wider life decisions (such as a decision not to have children, or not to embark on a particular career because of the incorrect belief that they [or their children] will become unwell later in life).

Historically (i.e., before the advent of massively parallel sequencing technologies), ascertainment of rare genetic variants was expensive, time consuming, and not widely available to nephrologists or other non-geneticists, so it was limited to those cases where there was a high prior probability of underlying Mendelian disease caused by a change in the gene (or few genes) analyzed. This meant that observation of a novel variant in such a targeted test performed in such a high-risk individual could be clinically consequential. Since the precipitous drop in sequencing cost and now widespread availability of large gene panels or even whole-exome sequencing in individual patients (termed "genomic testing"), the prior likelihood of a change seen in any one gene actually being the cause of a patient's disease has dropped. This is compounded by the fact that it is increasingly recognized that rare genetic variants can be associated with nonsyndromic (e.g., renal-limited) forms of diseases previously regarded as distinct syndromes. One example is the *LMX1B* variants associated with proteinuric kidney disease and focal segmental glomerulosclerosis (FSGS) in the absence of abnormal nail or patella development[2] that defined the Nail-patella syndrome, in which variants in this gene were originally identified. Together, this means that the previous situation where genetic tests would only be used after a clinical diagnosis of a specific syndrome or disease had been made is now being replaced by the paradigm in which larger numbers of patients lacking a specific clinical diagnosis are now undergoing large-scale genomic analysis. This has yielded a key benefit of securing an actionable molecular diagnosis in many more patients (as evidenced by the 15%–20% frequency of genetic diagnoses reported in patients with previously unexplained nephropathies undergoing exome sequencing[3]). But, as Bayes theorem indicates, the lower prior probability of a particular gene being responsible in patients undergoing testing has a dramatic effect on the posterior probability (and hence positive predictive value) of an identified genetic variant being the cause of a patient's disease. This mandates, more than ever before, that genetic variants identified in a patient are rigorously appraised to ensure that the actionability of each variant is commensurate with the evidence causally linking it to disease.

Understanding how to interpret (and therefore when to perform) genetic and genomic testing in the modern era therefore requires knowledge of not just which phenotypes or presentations can be associated with which types of genetic change (i.e., the **qualitative** question of whether a genetic change identified in a patient perturbs the function of a protein in a way consistent with their disease), but it also requires a **quantitative** understanding of how strongly a given type of genetic change is related to a disease (i.e., what is the absolute risk of the disease attributable to the genetic change identified in a patient). In communicating genetic test results to a patient, the clinician therefore needs to understand what information, for the variant in question, constitutes strong enough data on which to base potential highly consequential clinical and life decisions. This chapter therefore addresses the types and distributions of genetic variants found in humans; the clinical indicators of, and biological mechanisms causing Mendelian and non-Mendelian disease, as well as the sources of evidence linking genetic variation with disease. Finally, specific clinical presentations encountered by nephrologists are discussed to illustrate how this knowledge can inform clinical practice (Box 44.1).

TYPES OF GENETIC VARIATION

Most cells in the human body contain two copies of almost the entire genome of about 3 billion DNA nucleotides, around 1% to 2% of which encodes some 20,000 protein-coding genes and the rest comprising extensive noncoding sequences. Any two human beings differ from each other by about 0.1% of their genomes, and these differences are termed genetic variants, each of which has two or more versions (termed alleles) and therefore represent differences from the notional reference (or consensus) sequence of the human genome. These differences can be of several different types:

1. **Single nucleotide variants** (SNVs), which each comprise a single base-pair change. An estimated 84 million SNVs have been observed in humans. When they are known to be common (by convention occurring at a frequency exceeding 1% of a sufficiently large and diverse sample set of the global human population), they are termed single nucleotide polymorphisms (SNPs). They are reliably detectable using Sanger or massively parallel sequencing technologies or by SNP chips that can genotype millions of SNPs in an individual relatively cheaply.
2. **Insertions and deletions** (indels), which are small insertions or deletions ranging from 1 to around 50 base pairs depending on the definition used. Around 3.6 million indels have been reported in humans. These variants are also easily detectable using most sequencing technologies, and small indels can be genotyped using SNP chips.
3. **Structural variants** (SVs), which are changes that involve larger DNA segments (from around 50 up to millions of base pairs) that may be deleted, duplicated, inverted, or translocated. An estimated 60,000 SVs have been documented—some easily detected on classical karyotyping (e.g., trisomy 21) and some only detectable with recently developed long-read sequencing technologies. The term copy number variant (CNV) is a type of SV in which sections of the genome are deleted or duplicated, sometimes many times. There are estimated to be tens of thousands of CNVs, which are detectable using specific classical genetic assays (such as comparative genomic hybridization arrays or multiplex ligation-dependent probe amplification assays) and can also often be genotyped in an individual using many massive parallel sequencing approaches (including panel, whole-exome and whole-genome sequencing assays, provided read-depth, i.e., the number of times a particular sequence has been "read" by the assay, is sufficient) and some SNP chip platforms. They are usually not detectable by Sanger sequencing. Other, more complex SNVs (such as inversions and translocations) are not usually definable in an individual except using whole-genome sequencing: capture of (usually intergenic or deep intronic) breakpoints can allow reconstruction of the entire variant sequence using short-read WGS, or long-read sequencing can allow the full sequence of even extremely complex SVs to be read directly.
4. **Microsatellites** (or short tandem repeats, STRs) are repetitive sequences of usually two to six base pairs in length that are repeated a variable number of times. There are

Box 44.1 Genetic and Genomic Testing

Since the 1980s, polymerase chain reaction (PCR) and Sanger sequencing have been used to reveal the precise genetic sequence of small regions of a gene or locus (for instance, a single exon). This technique uses specific primers flanking a genomic region of (typically) a few hundred base pairs in length to amplify this segment (called an amplicon) chemically. This approach therefore usually requires a specific reaction to be developed for each exon of each gene to be amplified, with subsequent chemical termination of (again chemical) replication of the amplicon with randomly incorporated fluorescently labeled di-deoxynucleotide triphosphate molecules with a different excitation wavelength corresponding to each of the four types of DNA bases. These amplification products can then be separated electrophoretically, and the sequence can be read by the order (by size) of the relevant fluorescent labels. While cheap, readily accessible, and highly accurate, this approach (termed Sanger sequencing) is not easily scalable to allow sequencing of large numbers of exons or large genomic regions, so it becomes prohibitively costly when numerous exons or genes are to be sequenced. It is also not an effective way to ascertain structural variants (except for biallelic deletions, in which the amplification step will fail due to absence of the target sequence). Techniques to detect structural variants include karyotyping (sensitive only for chromosomal-scale abnormalities visible microscopically), comparative genomic hybridization, and multiplex ligation-dependent probe amplification (MLPA, which use quantitative hybridization or amplification of fluorescently labeled probes to detect variations in copy number across a number of genomic loci).

Genetic sequencing costs dropped substantially with the development of massively parallel or "next-generation" sequencing. There are a range of different implementations of this technology, which usually involve fragmentation of the DNA sample, attachment of adapters to the ends of the fragments to allow amplification of each fragment chemically (either fixed to a surface or in an emulsion). Fluorescent nucleotides are then added sequentially, and their incorporation is captured optically (generating a "read" for each fragment). Since millions of reads are captured in parallel, this generates a large amount of data from which the original genomic sequence can be inferred by assembly bioinformatically, either aligning them to a reference sequence or assembling them *de novo*. Massive parallel sequencing can be used following enrichment (e.g., with a library of all known exons, in whole-exome sequencing, or of a specified region in a targeted panel). In the absence of an enrichment step, the whole genome can be sequenced, but clearly more reads are needed to allow capture of more genomic loci.

Table 44.1 Variant Frequency and Typical Numbers in an Individual (data from 1000 Genomes and ExAC projects).[145,146]

Frequency Category	Typical Number in an Individual
>5% (common)	300,000-500,000
0.5%-5% (uncommon, or low frequency)	100,000-200,000
0.01%-0.5% (rare)	50,000-100,000
<0.01% (very rare or private)	>1,000,000

estimated to be hundreds of thousands of STRs in the human genome, and because of their highly polymorphic nature, they are often used for forensic, genealogic, or family-based studies.

Although most disease-associated genetic variants have a direct effect on protein production or function (i.e., the DNA change is within or intersects with a protein-coding gene), there are a growing number of examples of how variation in nonprotein coding regions of the genome can affect human biology, and as more human genomes have been sequenced in ever-larger cohorts, quantitative data are now emerging that reveal how these types of variants at each genomic locus can cause or contribute to disease. These often affect the quantity of protein produced (e.g., by affecting gene expression [as mRNA], microRNA-encoding genes, or genomic sequences that affect splicing of protein-encoding genes).

THE ALLELE FREQUENCY SPECTRUM IN HUMAN POPULATIONS

The frequency of the less common (or minor) allele of each variant can range between 0 and 0.5, with the number of variants in a typical person shown in (Table 44.1). However, the number of variants in each frequency category shows marked variation across different human populations, with the number of rare variants per individual being notably higher among those with recent African ancestry.[4]

Looked at across a population (Fig. 44.1), the vast majority of variants that can be detected by genome sequencing are ultrarare or private (i.e., occur in a single individual).

MENDELIAN DISEASES: INHERITANCE AND MECHANISMS

In nearly all human cells, the DNA carrying genetic information is organized into 22 pairs of autosomal chromosomes (autosomes), a pair of sex chromosomes, and numerous copies of the small mitochondrial genome. Normally, one copy of each autosome is inherited from each parent. Additionally, humans typically inherit one copy of their mother's X chromosome, and either their father's Y chromosome (in males) or their father's X chromosome (in females). Many copies of the mitochondrial genome are present in the egg from which each human originates, resulting in all mitochondrial genomes in an individual being maternally inherited. Thus each healthy human starts from a single cell with two copies (alleles) of each autosomal gene, one or two alleles of each X-linked gene (in males and females, respectively), one allele of each Y-chromosome gene (in males), and many copies of each mitochondrial gene.

Mendelian (or monogenic) diseases are defined as those in which a change in one or both copies of a single gene results in a high likelihood of a disorder that is otherwise rare. Such diseases can exhibit a Mendelian pattern of inheritance that is sometimes (although not always) readily ascertained from family history. However, since not all genetic changes exhibit complete penetrance[a]; some traits exhibit variable expressivity[b]; and in some families information regarding occurrence of the disease phenotype for individual members may be incomplete, inaccessible, or uncertain, it is not always possible to determine clinically what the mode of inheritance is, or indeed if a Mendelian trait is present at all. Where it can be ascertained, the inheritance pattern is determined by whether the genetic change causing the disease is located on an autosome, a sex chromosome, or within the mitochondrial genome.

AUTOSOMAL RECESSIVE DISEASES

In autosomal recessive diseases, the presence of two defective and no normal alleles of a relevant gene is usually necessary and sufficient to cause disease, and in general, it is deficiency (complete or near complete) of the normally functioning protein that leads to disease. This may be due to the inheritance of two identical defective alleles, called homozygosity, or two heterozygous alleles, termed compound heterozygosity when the alleles are *in trans* (i.e., there is a fault in both copies of the gene). Because a single normal copy of the gene is usually enough to prevent disease manifestations, carrier parents may be completely healthy, so the family history may reveal no affected relatives, affected siblings with healthy parents, or more distantly related affected relatives. Because defective alleles are rare in most populations, the occurrence of two defective alleles (one inherited from each parent) in an individual is usually a rare event. However, if the parents are related (e.g., cousins) or from the same genetically isolated population, the likelihood of homozygosity is increased because such parents are more likely to share a recent common ancestor from whom both may have inherited the same rare allele.[5,6]

Where biallelic variants are detected, inference or demonstration that they substantially reduce protein function or abundance can be sufficient to infer causation. Inference of deficiency can be made by observing that the variant(s) substantially shorten the length of any translated protein (e.g., by converting an amino acid codon into a stop codon, sometimes termed a nonsense mutation); by the variant being an indel that introduces a shift in the reading frame of the transcript (a frameshift mutation); demonstration that

[a]Penetrance is the proportion of individuals with a particular genotype (type of genetic change or changes) that exhibit the relevant phenotype.

[b]Variable expressivity refers (qualitatively) to the different manifestations (phenotypes) that can be exhibited by individuals with a particular genotype.

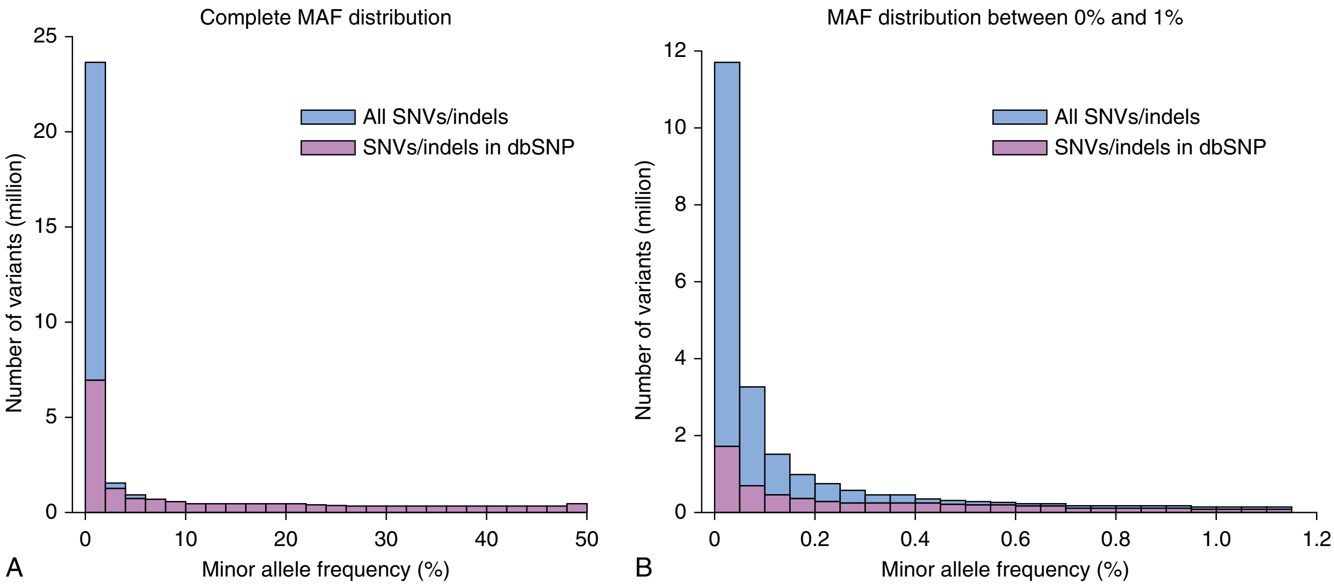

Fig. 44.1 The minor allele frequency (MAF) distribution in a European population. (Reproduced from Ameur A, Dahlberg J, Olason P, et al. SweGen: a whole-genome data resource of genetic variability in a cross-section of the Swedish population. *Eur J Hum Genet*. 2017;25(11):1253–1260.)

the variant affects splicing of the transcript in a way likely to substantially affect protein function (a splicing mutation); or the variant is a SV that disrupts the gene in such a way as to prevent production of the normal protein (e.g., deleting part or all of it). Where the variant introduces a change in the predicted protein from one (or a small number) of amino acids into a different amino acid(s) without affecting the length of the protein (termed a missense mutation or small in-frame indel), additional evidence is usually needed before such a variant can be inferred to abrogate protein function. This additional evidence might include an assay demonstrating that the protein level in the blood (or other relevant tissue) of the patient is substantially reduced or if there is an established biochemical assay for the function of the protein showing this is substantially reduced (e.g., assays for the von Willebrand Factor cleavage activity of ADAMTS13 used to diagnose congenital thrombotic thrombocytopenic purpura). Sometimes, the loss of function can also be inferred (e.g., if the variant changes an evolutionary highly conserved amino acid or alters a domain that is known to be critical for protein function). Various computer algorithms that generate predictions about the deleteriousness of the variant based on such criteria exist, but such *in silico* evidence is obviously less reliable than a readout of protein function from a clinical assay. This is reflected in the variant annotation criteria (see "Genetic Variant Interpretation in Mendelian Diseases" later), where computer predictions only provide "supportive" evidence, whereas a clinical assay counts as "strong" evidence.

If both parents are found to be heterozygous for an allele that causes a recessive disease, the risk to each of their future offspring of having the disease is 25% since each would need to inherit a defective allele from both parents to be affected. This knowledge might have implications for parents of an affected child considering having more children. In most (but not all) recessive diseases, heterozygous relatives are at such low risk of disease that they are not offered surveillance or precluded from donating a kidney. One important exception to this is in autosomal recessive Alport syndrome, where heterozygous relatives should undergo thorough clinical appraisal and consider carefully the pros and cons of kidney donation, especially with reference to their age at the time of donation (see later).

AUTOSOMAL DOMINANT DISEASES

In autosomal dominant disorders, a single defective allele can cause the disease, even if a normal allele is present on the sister chromosome. Consequently, each child of an affected individual has a 50% chance of inheriting the defective allele. This transmission risk is the same for both sexes and affected individuals often appear in successive generations. However, many autosomal-dominant disorders exhibit incomplete penetrance, meaning the probability of showing symptoms despite having a causative allele is <100%. Factors influencing penetrance include the precise mutation present in a family, environmental exposures, and genetic variation in other genes (genetic modifiers). Incomplete penetrance can make the disease appear to skip generations, and distinguishing autosomal-dominant disorders from X-linked recessive disorders can sometimes rely on identifying male-to-male transmission when eliciting the family history (making identification of such transmissions extremely valuable when taking a family history).

For an autosomal-dominant disorder to be passed to the next generation, the defective allele must allow survival to adulthood and fitness to reproduce. This could be due to incomplete penetrance or the disorder, affecting survival later in life. *De novo* mutations account for a significant proportion of cases, especially in potentially severe childhood-onset disorders like WT1 disorders Townes-Brocks syndrome, and Tuberous Sclerosis Complex that often occur in children without a family history. Furthermore, large, repetitive genes such as *PKD1* are especially prone to such mutational events, explaining why autosomal dominant polycystic kidney disease is so common (in all human populations) and frequently occurs in individuals with no family history. Dominant disorders can also first appear in a family due to somatic mosaicism in a parent, where the mutation arises during embryogenesis and is present in the gonadal tissue but not in the affected organ, making the parent healthy but capable of passing the mutation to offspring. This implies a nonzero risk of future siblings being affected, even if both parents are unaffected.

Because they occur alongside a normal allele, the mechanisms by which dominant alleles cause disease can be more complex than simple deficiency of a protein and can be categorized into four groups:

1. **Haploinsufficiency**, where the presence of a single loss-of-function allele (in the presence of a normal allele on the other chromosome) results in not enough protein being made to avoid disease. This is a relatively uncommon mechanism because in most situations, transcriptional upregulation occurs, resulting in more copies of the (normal and faulty) mRNA being produced, leading to (near) normal protein abundance. Haploinsufficiency can occur when the quantity of an enzyme is critical for controlling a biological process. For instance, in PHD2 erythrocytosis, even a small decrease in the PHD2 enzyme level lowers the efficiency of degradation of hypoxia-inducible factor (HIF) 2-α, a transcription factor that promotes erythropoietin production in the kidneys, resulting in elevated hematocrit and erythrocytosis.[7] Haploinsufficiency is also the mechanism in many disorders due to variants in genes affecting transcription factors (e.g., HNF1B), as dosage of transcription factors appears critical for many processes,
2. **Gain of function,** where the change in the gene, rather than reducing the function of the relevant protein leads to its increase (e.g., by impairing its normal degradation). An example of such an *activating mutation* is seen in HIF2-α erythrocytosis, where specific amino acid substitutions result in a protein that is more resistant to the action of PHD2 and hence oxygen-dependent degradation than is the wild-type allele, resulting in increased HIF2-α abundance, elevated hematocrit, erythrocytosis, and pulmonary hypertension.[8] Other gain-of-function mechanisms include *mistargeting mutation,* such as a change in *EHHADH* that introduces an abnormal mitochondrial targeting motif, leading to increased abundance of this (usually peroxisomally located) protein in mitochondria, where it impairs oxidative phosphorylation in tubular cells, resulting in renal Fanconi syndrome.[9] A third type of gain-of-function mechanism is where a change in the amino acid structure of the protein confers on the protein a *new function* that causes disease. An example of this is the presence of a certain single amino acid substitution in the fibrinogen protein, which confers on the protein a tendency to form amyloid fibrils that deposit in tissues, causing disease.[10] Clearly, in such a situation, whole-gene deletion or truncating variants associated with nonsense-mediated decay would not be predicted to be pathogenic, and only highly specific missense variants (that have the required gain-of-function effect) would be expected to cause the disease. This is important to consider when trying to interpret a given variant, as the commonly used computer algorithms are designed to predict deleteriousness, but not a specific gain of function. Thus these algorithms should not be used in a disorder with a gain-of-function mechanism because that could lead to erroneous variant interpretation.
3. **Dominant negative,** where the variant causes a faulty protein to be produced that not only has impaired function but also inhibits the function of the wild-type protein. An example of this includes certain growth hormone receptor mutations that eliminate the intracellular domains of the protein while leaving the extracellular and transmembrane domains intact. The resulting protein is missing the segments necessary for degradation and intracellular signaling but can still bind to growth hormone (its ligand) and form dimers with the wild-type allele. Consequently, it accumulates and competes with the normal protein, interfering with growth hormone signaling and leading to congenital growth hormone insensitivity.[11]
4. **Somatic second hit,** where loss of the other (functional) copy of the gene in a single cell due to a somatic mutational event renders that cell unable to make the relevant protein because it now has no functioning copy of the gene. This explains many autosomal dominant cancer syndromes, where lack of a component of cellular machinery that normally controls cell division makes the cell much more prone to uncontrolled proliferation. According to Knudson's two-hit hypothesis, in individuals with two functional alleles of the gene in their germline genome, two separate mutational events in a cell are required to render the cell deficient in the relevant protein and hence prone to become cancerous. In those individuals who inherit one faulty copy (which is therefore present in all their cells) from a similarly affected parent, only a single somatic hit is required, making their exposure to the risk of the relevant disease orders of magnitude higher than those lacking such susceptibility. An example of this mechanism is von Hippel Lindau disease, in which loss of both copies of the *VHL* gene in a cell renders that the cell is unable to make the VHL protein and therefore unable to break down HIF (even in the presence of normal oxygen levels). This results in the widespread activation of a host of cellular, metabolic, and physiologic responses to hypoxia that strongly predispose the cell to developing into a clear cell renal cell carcinoma.[12,13] Regular monitoring to detect early renal cancers can significantly improve prognosis in such individuals, making predictive genetic testing particularly valuable for at-risk relatives of affected individuals. Diseases caused by a somatic second-hit mechanism also include nonmalignant conditions such as autosomal dominant polycystic kidney disease (where the faster accumulation of cysts in *PKD1*-associated compared with *PKD2*-associated disease may be attributable to the larger size, and hence greater susceptibility to mutational events, of the *PKD1* gene). Somatic second-hit–mediated phenotypes may not appear until adolescence or adulthood, tend to be multifocal (arising as separate abnormalities at different times and locations within a susceptible organ), and can vary greatly in age of onset and severity within a family. This variability likely stems from the random occurrence of somatic mutations, variability in genes responsible for detecting and repairing mutations during cell division, and the influence of environmental factors (such as

smoking) on the rate of somatic mutations. This latter consideration makes avoidance of carcinogen exposure during life by individuals with a somatic second-hit–mediated disease particularly advisable.

X-LINKED RECESSIVE DISEASES

In X-linked recessive diseases, the severity or manifestations of disease are usually different in males and females. In males, the presence of a single faulty copy of a gene on the X chromosome implies the absence of a normal copy of the gene, and hence loss-of-function alleles can be inferred to be disease causing by resulting in deficiency of the protein. In females the situation is more complicated, as inactivation of one (usually at random) copy of the X chromosome in each cell means that they are essentially mosaic for the normal and faulty alleles. Whether this mosaicism has clinical consequences depends on the gene in question. In addition, it is possible that heterozygosity for a faulty X-linked gene can have haploinsufficiency or even gain-of-function effect. Therefore it is not surprising that females with either Fabry disease or X-linked Alport syndrome are at increased risk of kidney disease, although this tends to occur later in life than it does in males with these disorders.

MITOCHONDRIALLY INHERITED DISEASES

The human mitochondrial genome has around 16,500 base pairs (compared with 3 billion for the rest of the genome) and only 37 genes, encoding only a small proportion of the components these organelles need to function, with the rest being encoded in nuclear DNA. This means that only a small proportion of mitochondrial disorders are mitochondrially inherited. Since each sperm is thought to contain fewer than 5 mitochondria (which are believed to be devoid of intact mitochondrial DNA), compared with around 200,000 within each egg, there is exclusive maternal inheritance of mitochondrial variants.[14] Mitochondrial DNA is circular and present in multiple copies in each mitochondrion. If all the copies in all mitochondria in the egg have a variant, it will be uniformly present in all the cells of the organism (termed homoplasmy). More commonly, a genetic variant will be present in only some copies or some mitochondria (termed heteroplasmy), in which situation, as cells divide in a developing embryo, the proportion of defective mitochondrial genomes can vary across different tissues and organs. This implies that while all offspring of an affected mother are at risk of the disease, each egg contains a unique proportion of defective mitochondrial genomes that are randomly distributed in the developing embryo. Consequently, the manifestations and severity of the disease can differ significantly among affected individuals within the same family. A typical family history of a mitochondrially inherited disorder will show a wide range of severity and symptoms among family members and tellingly will never include transmission from affected males to their offspring.

Clearly, knowledge of the likely mechanism by which changes in a gene cause disease, as well as the types of genetic changes that can mediate these mechanisms, is critical for interpretation of genetic test results: Inferring pathogenicity of a deletion or early truncating mutation seen in a gene associated with disease via a gain-of-function mechanism is likely to be erroneous. While some monogenic diseases are highly penetrant, other conditions, particularly autosomal dominant diseases with onset later in life, can exhibit incomplete penetrance, with some individuals with the disease-associated genetic change(s) never becoming aware that they have the condition. In general, for autosomal dominant conditions in societies in which family size is small (typical sibships of 2–4 individuals), once the penetrance falls below 50%, the observation of a Mendelian pattern of inheritance becomes unusual and is even more so where penetrance is below 20%. For some diseases, the penetrance can be critically sensitive to the environment individuals are exposed to or to the presence of (non-Mendelian) genetic factors elsewhere in the genome, so a given genetic change may exhibit Mendelian inheritance in some, but not all, families it is present in. This illustrates that there is no clear-cut definition of what constitutes a monogenic disease allele versus a risk factor and implies that, when counseling patients, clinicians need an understanding of genetics that is broader than the classical concepts described by Mendel.

GENETIC VARIANT INTERPRETATION IN MENDELIAN DISEASES

When a rare genetic variant is identified in an individual in a gene known to have been reliably associated with a Mendelian disease, in most testing systems a molecular genetics specialist, sometimes in consultation with the clinician, will endeavor to interpret the finding. Determining whether the finding can be reliably inferred to explain disease in that patient is therefore clinically actionable. As detailed earlier, the extremely consequential (and often predictive) uses of this information mandate a high degree of certainty before such variants are reportable as diagnostic findings. For clinicians involved in genetic testing, the overriding consideration is not the specificity of the test (specificity is the degree of certainty the laboratory has that a variant identified is pathogenic, as expressed by the proportion of such variants identified by the laboratory that are, in fact, disease causing), but rather it is the **positive predictive value** of the test (which is the probability of the patient's disease being explained by the variant identified), *which is critically dependent on the prior probability that the patient's disease will be explained by such a molecular finding*. As for any medical test, the positive predictive value is the ratio of true positives to the sum of the true and false positives identified by the test and can be calculated using the specificity and sensitivity of the test, as well as the prevalence (P) of the condition *in the group of patients subjected to the test* using the formula:

$$\text{Positive predictive value} = \text{Sensitivity} \cdot \text{P} / (\text{Sensitivity} \cdot \text{P} + [1 - \text{Specificity}] \cdot [1 - \text{P}])$$

Because genetic tests have, at best, a sensitivity of around 85% (some patients will have deep intronic or SVs presently difficult to detect), the relationship between positive predictive value and specificity is affected by the value of P, as illustrated in Fig. 44.2.

Therefore where there is a high prevalence (>60%, say) of the condition in the population being tested (e.g., polycystic kidney disease in individuals with enlarged cystic kidneys and liver cysts), even a 90% certainty that a variant

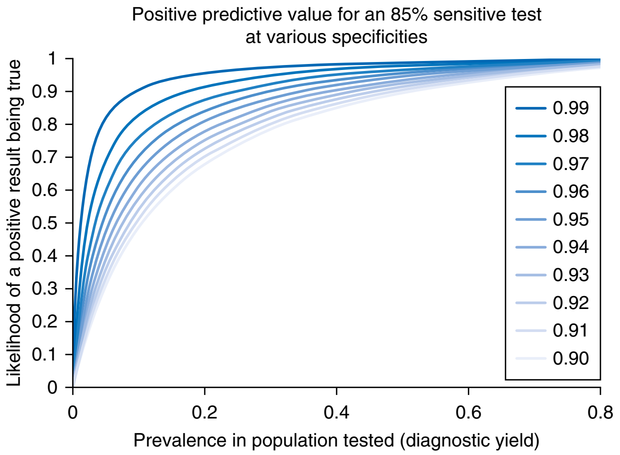

Fig. 44.2 Positive predictive value for an 85% sensitive test according to prior probability, or prevalence of the disease in the population in which the test is applied, with each line representing a specificity between 0.90 and 0.99.

causes a disease is associated with a posterior probability of such a variant causing disease in a patient of >90% and so is actionable. In contrast, where the diagnostic yield in a population is below 20% (which it might be if testing is routinely performed in children with nonsyndromic, nonfamilial developmental abnormalities of the urinary tract or steroid-resistant nephrotic syndrome), the posterior probability for a variant for which there is deemed to be a 90% chance of being disease causing is below 70% and therefore unlikely to be clinically actionable. Clearly, if the laboratory is >99% certain the variant is disease causing, such a result can be clinically actionable if the prior probability is around 20%, but below a prior probability of 5%, only variants whose pathogenicity is certain can be safely used to inform consequential clinical or other decisions. Intuitively, this can be explained by considering a specificity of 99% as equivalent to reporting one false positive in every hundred samples tested. If, among 100 samples tested there are only 5 true positives likely to be identified, then of the 6 positive results reported by the laboratory, only approximately 5 (83%) of them will be true positives. Therefore for each patient receiving such a report, there is a 1:6 chance of assigning an incorrect molecular diagnosis. For a clinician who only seldom orders a test, it may feel ethically difficult to ignore the presence of a rare or otherwise suspicious variant for which the level of certainty it is disease-causing is <99% in an individual patient. However, it is important, for the reasons outlined earlier, to consider in this situation *the distribution of test results across the population of patients being tested.* A strong argument can therefore be made in some circumstances for not releasing information about variants for which there is uncertainty of pathogenicity.

Of course, this can also work the other way: In a disease with a specific phenotype and precise clinical diagnosis and a single genetic etiology, even uncertain variants identified in this disease gene are likely to be causative, if found in a patient with this phenotype (see also “Tubulopathies” later). This highlights the importance of close collaboration between the clinician requesting the test and the genetic scientist carrying out the test. A genetic diagnosis is not an invariable result that can be obtained in isolation, such as by simply sending a blood sample to a laboratory with the request to identify the cause of the patient’s kidney disease. Rather, identified variants must be carefully interpreted by considering multiple aspects, a key one being the clinical phenotype and how well it fits with the identified variants. In addition, a solid understanding of the disease gene is important, as in some genes pathogenic variants typically cluster in specific regions (e.g., *WT1*, exons 8 and 9[15]) or even polymorphisms can be pathogenic if *in trans* with specific other variants (e.g., p.Arg229Gln in *NPHS2*[16]).

This need for careful interpretation can result in conflicting assessments between laboratories. To improve and standardize this, the American College of Medical Genetics and Genomics (ACMG) and the Association for Molecular Pathology (AMP) issued guidance on how to interpret genetic variants that were well summarized in a joint ACMG/AMP consensus recommendation in 2015.[17] They proposed 5 classifications of variants with only classes 4 and 5 (i.e., pathogenic and likely pathogenic) being clinically actionable:

1. Benign
2. Likely benign
3. Variant of uncertain significance (VUS)
4. Likely pathogenic
5. Pathogenic

In addition, they also specified the criteria on which variants could be assigned to each class, in two key figures comprising the evidence framework (Table 44.2) and how the components of evidence can be integrated to assign pathogenicity class (Fig. 44.3).

Although requiring refinement (particularly to address ambiguities, such as what constitutes a “well established functional study”), this has provided the international community with an invaluable framework on which to base variant interpretation, taking into account what is known or inferred (computationally or experimentally) about the effect of the variant on protein structure or function; frequency of the variant in population databases; and any previous reports of that variant in disease, taking into account that any previous reports of pathogenicity should have followed similarly robust pathogenicity assessment. It should be noted that, although not strictly defined, in the 2015 consensus statement (following a survey of the community during an ACMG open forum) the term “likely pathogenic” was proposed to mean >90% certainty of a variant being disease causing. Whether this level of certainty remains optimal given the wider usage of genomic testing in the current era remains to be determined. However, the likelihood of being pathogenic as defined in the ACMG criteria is only estimated and may reflect conservative assumptions. For this reason, retrospective studies that aim to correlate the clinical phenotype with a given genetic diagnosis are important to get a better idea of the true specificity of the test. It is also important to emphasize that Mendelian disease classification is not a clinically useful way to classify non-Mendelian genetic variants (e.g., a heterozygous deleterious variant in a gene associated with a recessive disease) because there can be a high statistical degree of certainty that a non-Mendelian genetic variant has a (sometimes small) effect on disease risk—meaning that such a variant may be known to contribute to disease pathogenicity, without it being “pathogenic” in the sense that it is the clinically actionable cause of a patient’s or family’s

Table 44.2 Rules for Combining Criteria to Classify Sequence Variants[17]

Pathogenic
1. 1 Very Strong (PVS1) *AND*
a. ≥1 Strong (PS1-PS4) *OR*
b. ≥2 Moderate (PM1-PM6) *OR*
c. 1 Moderate (PM1-PM6) and 1 supporting (PP1-PP5) *OR*
d. ≥2 Supporting (PP1-PP5)
2. ≥2 Strong (PS1-PS4) OR
3. 1 Strong (PS1-PS4) AND
a. ≥3 Moderate (PM1-PM6) *OR*
b. 2 Moderate (PM1-PM6) *AND* ≥2 Supporting (PP1-PP5) *OR*
c. 1 Moderate (PM1-PM6) *AND* ≥4 Supporting (PP1-PP5)
Likely Pathogenic
1. 1 Very Strong (PVS1) *AND* 1 Moderate (PM1-PM6) *OR*
2. 1 Strong (PS1-PS4) *AND* 1-2 Moderate (PM1-PM6) *OR*
3. 1 Strong (PS1-PS4) *AND* ≥2 Supporting (PP1-PP5) *OR*
4. ≥3 Moderate (PM1-PM6) *OR*
5. 2 Moderate (PM1-PM6) *AND* ≥2 Supporting (PP1-PP5) *OR*
6. 1 Moderate (PM1-PM6) *AND* ≥4 Supporting (PP1-PP5)
Benign
1. 1 Stand-Alone (BA1) *OR*
2. ≥2 Strong (BS1-BS4)
Likely Benign
1. 1 Strong (BS1-BS4) and 1 Supporting (BP1-BP7) *OR*
2. ≥2 Supporting (BP1-BP7)

*Variants should be classified as Uncertain Significance if other criteria are unmet or the criteria for benign and pathogenic are contradictory.

disease. Clinical genetics programs have established resources aimed at classifying gene-disease relationships for clinical use (e.g., the crowd-sourced Genomics England PanelApp, https://panelapp.genomicsengland.co.uk). A U.S. National Institute of Health–sponsored international effort, called ClinGen, is under way aiming to build a central resource that defines the clinical relevance of genes and variants for use in precision medicine and research (https://www.clinicalgenome.org).[18] From this it becomes obvious that a genetic diagnosis can change over time: A variant previously considered causative may be reconsidered because of the discovery of an unrealistically high allele frequency in the general population (see criterion BS1 in Table 44.2). Indeed, it has been estimated that up to a quarter of published "disease-causing" variants may be erroneously considered pathogenic.[19,20] Conversely, a variant previously considered to be of uncertain significance may be reassessed as (likely) pathogenic due to its identification in multiple other patients with the same phenotype or because of new mechanistic insights in how the variant affects protein function.

NON-MENDELIAN DISEASE GENETICS

While a genetic change that conveys a high likelihood of a specific disease can lead to a Mendelian pattern of inheritance within a family, most genetic variants each have a much smaller (or no) effect on disease risk. In such a situation the effect of such a variant can only be reliably detected using an association study, where the frequency of an allele among a group of people with the disease (cases) is different from the frequency among those lacking (or not enriched for) it (controls). The power of such a study to detect an effect on disease risk of an allele is dependent on three key factors: the number of cases and controls in the study; the magnitude of the effect on disease risk of the genetic variants present in the population; and, critically, the frequency of the allele that confers altered risk of the disease in the population under study. To avoid issues of confounding, it is important that the cases and controls are well matched according to ancestry, relatedness, and environmental exposures because otherwise alleles that have a different frequency in different ancestral populations might be mistakenly inferred to be associated with the disease (a phenomenon called population stratification). Carefully designed association studies that use SNP chips to genotype hundreds of thousands, or even millions, of (mostly common) SNPs have therefore proved a sensitive way to detect common alleles in a population that affect risk of disease. Because alleles close to each other in the genome tend to be inherited together (such a group is called a "haplotype" of variants that exhibit "linkage disequilibrium" with each other, or coinheritance more often than would be expected if they were transmitted independently by chance), the genotype of a small number of SNPs at a locus (genomic region) in an individual can allow imputation of a large number of nongenotyped variants around them using reference panels comprising whole-genome sequencing data from relatively small numbers of individuals. The effect of linkage disequilibrium between nearby alleles means that genotypes of nearby SNPs are not independent of each other. This has two important corollaries. First, it allows genome-wide association studies (GWASs) that can ascertain the effects of common and uncommon variants across the entire genome on the risk of a disease to be performed. Second, the number of independent tests performed in such GWASs, while large, is not the same as the number of SNPs genotyped or imputed, leading to a conventionally accepted threshold for statistical significance for such studies of $P < 5 \times 10^{-8}$. However, because rare, ultrarare, or private variants are, by virtue of their rarity, not usually inherited in multiple unrelated individuals on the same haplotype (otherwise they would be common), such variants cannot usually be imputed or tested in SNP chip–based GWASs and require reliable ascertainment of a genome sequence across large cohorts to be ascertained (such a study is sometimes termed a whole-genome sequence–based association study, or seqGWAS).

Because, for a GWAS of a given size, rarer alleles need to have greater effect on disease risk than commoner alleles to be detectable, GWASs are ideally suited (and were first deployed in) common diseases, where it was relatively easy to acquire relatively large samples. Such studies, first published in the early 2000s, elucidated unassailable evidence for the role of common variants in predisposing to the risk of common diseases.[21,22] In addition, over time, these and similar landmark studies were able to confirm some and refute many previous candidate gene–based association

	Benign		Pathogenic			
	Strong	**Supporting**	**Supporting**	**Moderate**	**Strong**	**Very strong**
Population data	MAF is too high for disorder *BA1/BS1* OR observation in controls inconsistent with disease penetrance *BS2*			Absent in population databases *PM2*	Prevalence in affecteds statistically increased over controls *PS4*	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene/gene product *BP4* Missense in gene where only truncating cause disease *BP1* Silent variant with non-predicted splice impact *BP7*	Multiple lines of computational evidence support a deleterious effect on the gene/gene product *PP3*	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before *PM5* Protein length changing variant *PM4*	Same amino acid change as an established pathogenic variant *PS1*	Predicted null variant in a gene where LOF is a known mechanism of disease *PVS1*
Functional data	Well-established functional studies show no deleterious effect *BS3*		Missense in gene with low rate of benign missense variants and pathogenic missenses common *PP2*	Mutational hot spot or well-studied functional domain without benign variation *PM1*	Well-established functional studies show a deleterious effect *PS3*	
Segregation data	Nonsegregation with disease *BS4*		Cosegregation with disease in multiple affected family members *PP1*	Increased segregation data →	→	
***De novo* data**				*De novo* (without paternity and maternity confirmed) *PM6*	*De novo* (paternity and maternity confirmed) *PS2*	
Allelic data		Observed in *trans* with a dominant variant *BP2* Observed in *cis* with a pathogenic variant *BP2*		For recessive disorders, detected in *trans* with a pathogenic variant *PM3*		
Other database		Reputable source w/out shared data = benign *BP6*	Reputable source = pathogenic *PP5*			
Other data		Found in case with an alternate cause *BP5*	Patient's phenotype or FH highly specific for gene *PP4*			

Fig. 44.3 Evidence framework for assigning attribution of pathogenicity to variants.[17] *BP,* Benign supporting; *BS,* benign strong; *PM*, pathogenic moderate; *PP,* pathogenic supporting; *PS,* pathogenic strong; *PVS,* pathogenic very strong.

studies that had appeared in the scientific literature, some with detailed experimental support.[23-25] What allowed this critical reappraisal of the prior literature was the use of stringent statistical significance thresholds in combination with the unbiased genome-wide assessment of genetic risk factors that allowed appreciation of the null distribution and more reliable identification of those alleles lying outside it. Candidate gene studies, which reported alleles that reached notional statistical significance ($P < 0.05$) in analyses of a single gene, without publication of the numerous potentially testable (and sometimes tested) alleles at either the same locus or other genes investigated, typically lacked these controls and were therefore highly susceptible to type 1 error. Historically, such studies were sometimes accompanied by detailed functional assessments showing laboratory evidence that the alleles identified could (in a carefully controlled model or experimental setting) affect protein function. While no longer often cited, this literature (from the pre-GWAS era) is informative as to how ascertainment of genetic variation without a firm understanding of the null distribution can mislead scientists and prompt (sometimes many) highly sophisticated laboratory experiments that are based only on statistical noise and which are at times extremely difficult to interpret. Consequently, reports of genetic association studies now usually require similar levels of stringent statistical significance and genomic control (which can ascertain hidden population stratification and reveal the underlying null distribution of genetic variation across the genome) to GWASs if they are to be usefully incorporated into the scientific literature.

The identification, by unbiased GWASs, of specific alleles that convey risk of disease has been of enormous scientific value because identification of the relevant gene and biological mechanism linking the genetic variant to disease can reveal novel biological mechanisms contributing to disease, which, in some cases, can inform development of

Table 44.3 Examples of Genetic Associations Identified Using Genome-Wide Association Studies

Condition	Associated Loci (Represented By Nearest Gene)	Odds Ratio	Example Reference
IgA nephropathy	FCRL3, TNFSF4/18, CFH/CFHR1-3, REL, CD28, PF4V1, IRF4/DUSP22, LY86, HLA-DRA, DQB, DQA, DPA, DPB, DEFA1/4, LYN, ANXA3, TNFSF8/15, CARD9, REEP3, ZMIZ1, OVOL1/RELA, ETS1, IGH, ITGAM/ITGAX, IRF8, TNFSF12/13, TNFRSF13B, FCAR, LIF/OSM	1.12-1.33	26
IgA vasculitis	HLA-DQB, DRB	1.59	147
IC-MPGN/C3G	HLA-DQA1/DRB1	1.93	103
Membranous nephropathy	HLA-DQA/DRB, PLA2R1, IRF4, NKFB1	1.25-2.41[a]	148
Steroid-sensitive nephrotic syndrome	HLA-DQB, NPHS1, CALHM6, AHI1, TNFSF15, CLEC16A, CD28, BTC	0.46-2.04	149
PR3 vasculitis	HLA-DP, SERPINA1, ARGHAP18, PRTN3	0.53-1.55	35
MPO-vasculitis	HLA-DQ	0.65	35
Eosinophilic granulomatosis with polyangiitis	HLA-DQB, RNU7-106P	1.62-2.0[b]	150
Posterior urethral valves	TBX5, PTK7	0.4–7.0	151
Bladder exstrophy	LPHN2, EFNA1, SLC50A1, DPM3, KRTCAP2, ISL1, TRIM29, SYT1, PAWR, GOSR2, LINC01974, LINC01716, HMGB1P47, ISL1-DT	1.47-2.35	152
Urinary stone disease	ALPL, GCKR, DGKD, ABCG2, SCL34A1, KCNK5, SLC22A2, HIBADH, AQP1, POU2AF, DGKH, WDR72, UMOD, SCNN1B, BCAS, SOX9, GIPC1, CYP24A1, CLDN14, BCR	1.05-1.24	27

[a]Strong evidence of epistasis between *PLA2R1* and *HLA-DRB* (see text).

therapies. Examples of results of GWASs in specific renal diseases are shown in Table 44.3. Identification of associated genes has been of particular value in IgA nephropathy, in which numerous adaptive and innate immune genes were implicated, providing strong support for targeting the complement system therapeutically in the disease,[26] and in urinary stone disease, in which modulation of calcium sensing and vitamin D metabolism has been strongly implicated.[27] GWASs for reduced eGFR or CKD have identified at least 264 associated loci including common variants thought to affect *UMOD* expression.[28] It should be noted that, in most kidney diseases, the common variants accessible to study by GWASs explain only a small proportion of the observed heritability (that is the proportion of variation in a trait attributable to genetic factors). In addition, because the effect size of most associations discovered in this way is small, the identification of the presence of a common variant that is a risk factor for a disease in an individual patient has not usually been clinically actionable so testing for these (generally weak) genetic risk factors for disease is not usual in clinical practice.

In some diseases, however, the presence of multiple common genetic risk alleles (each conveying typically only a small risk individually), if aggregated across the genome in an individual, can have a strong enough effect on disease risk to have the potential to inform clinical practice.[29] The combined effect on disease risk of multiple alleles is termed, variously, a "genetic risk score," "polygenic risk score," or "polygenic score" depending on how they are calculated and, while none are in widespread clinical use in renal diseases at the time of writing, it is possible that such scores will be incorporated into clinical practice in nephrology in the future and, as shown later, their use in a research setting can be informative about the biology and epidemiology of (even rare) diseases.

ALLELE FREQUENCY AND DISEASE RISK

In terms of effect on the person in whom it is found, a variant can be silent (i.e., have no detectable effect on the organism), be harmless (i.e., have an effect that does not reduce fitness, such as on eye color), or confer risk of a disease. This risk may be small, moderate, or large. It is relatively unusual for common variants to carry a high risk of disease because high-risk alleles tend to be selected out by natural selection. An exception to this is "balancing selection," in which an allele that conveys a high risk of one disease also confers protection against a different (often infectious) disease. In this situation, seen in hemoglobinopathy- or thalassemia-causing alleles that protect from malaria, or certain *APOL1* alleles predisposing to glomerulopathy that confer protection from trypanosomiasis,[30,31] the disease-causing allele can become common in a population. Rare variants, because they are not shared by large numbers of individuals, may have arisen recently, so they can be associated with greater risk of disease (as they have not been subjected to selection pressure over many generations). This is why Mendelian disease alleles tend to be rare, with exceptions being the abovementioned hemoglobinopathies or thalassemias and some other (usually recessive) alleles that have become common over time in a population due to genetic drift.

The allele frequency spectrum and its association with disease risk spectrum have historically been conceptualized as summarized in Fig. 44.4.

However, this paradigm reflects the situation as it was in the 2000s, where rare alleles could not be ascertained

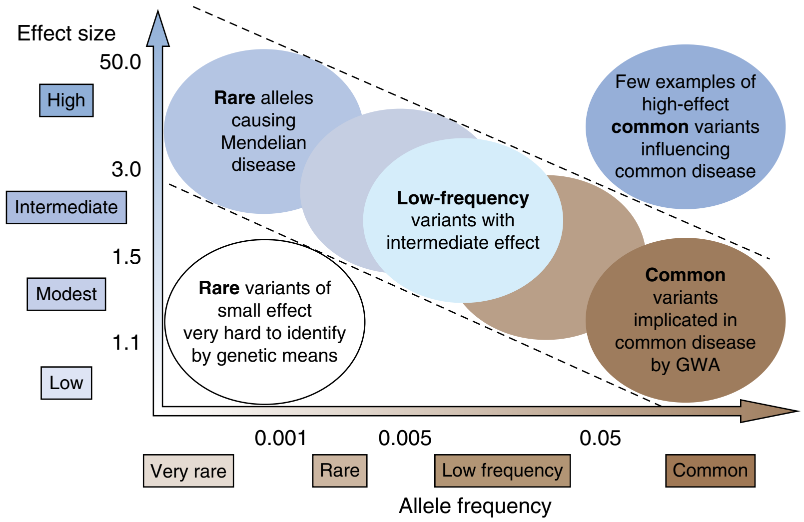

Fig. 44.4 Frequency and effect on disease risk of genetic variants, as perceived in 2010.[143]

in large cohorts of individuals and where there were few examples of common variants causing a large increase in risk of rare diseases (e.g., *APOL1* alleles predisposing to glomerulopathy). Subsequently, large biobank whole-genome sequencing endeavors such as the UK Biobank[32] and 100,000 Genomes Project[33] have become available and empiric evidence is mounting that rare variants can convey any magnitude of disease risk, from zero to very high effect size. Coupling these data with the knowledge that most genetic variation between individuals is composed of rare, ultrarare, or private genetic variants, it is almost self-evident that observation of a rare variant in a particular gene in an individual is not sufficient, by itself, to infer that the variant conveys a high enough risk of disease to be regarded as actionable as a cause of a Mendelian disease: It could have no effect on disease risk (i.e., be functionally silent or harmless) or it could perturb function of the relevant protein or biological system. However, even a reliable demonstration using laboratory systems (termed “functional characterization” of a variant) that the variant influences the abundance or function of the relevant protein, while consistent with the allele affecting disease susceptibility, is not always sufficient to reveal the *magnitude* of any effect on disease risk. To draw clinically actionable conclusions, additional, usually clinical or population-based evidence is needed, informed by understanding of the mechanism(s) linking variation in that gene with the disease.

INTERPRETATION OF GENETIC ASSOCIATIONS IN CLINICAL PRACTICE

In the early stages, GWASs were usually performed to study common diseases and the typical odds ratio of SNP alleles was identified (i.e., their effect on disease relative risk) was between 1.2 and 2.0.[22] As cohorts of people with less common and even rare diseases have been assembled since the 2000s, GWASs in important but rare renal diseases such as IgA nephropathy, membranous nephropathy, and vasculitis started to be published.[34-36] Because of the small size (typically comprising hundreds rather than thousands of cases), such studies were able to detect (initially at least) only those alleles that are both common in the population in which the studies were performed and which also had a relatively large effect size. One such GWAS, published in 2011, that included 556 patients with membranous nephropathy and 2338 unaffected individuals identified risk alleles at *PLA2R1* (the gene encoding the phospholipase A2 receptor—which had been recently reported to be the antigen implicated in the majority of those affected[37]) and certain human leukocyte antigen (HLA) alleles. The odds ratio for homozygosity of common risk alleles at both loci was a notable 78.5 (95% confidence interval 34.6–178.2),[36] which is orders of magnitude greater than that seen in most other GWAS. As well as providing genetic evidence for the immunologic mechanism driving this disease, this study therefore also proved that rare diseases were amenable to study using GWAS and that effect size (in terms of relative risk) of common alleles does not necessarily have to be small, meaning that relatively small-scale association studies can reveal genetic risk effects and that sometimes common variants could have a large effect on the relative risk of a rare disease. Importantly, even a large effect size of a variant (in terms of odds ratio or relative risk of disease) can still be consistent with only a small absolute risk of the disease in the presence of the variant, meaning that identification of the allele (or alleles) in an individual is not clinically actionable. In membranous nephropathy, which has an estimated incidence of 10 to 12 per million population per year,[38] even the 78.5-fold increase in disease risk among individuals who are homozygous for both risk alleles approximates to a predicted incidence of well below 1:1000. This is far below the level of risk at which medical interventions or surveillance could be helpfully delivered.

This illustrates that to determine whether identification of a variant, or type of variant, in an individual is clinically actionable, the key piece of information needed is the absolute risk of disease among individuals from that population (sharing similar environmental exposures) who harbor that variant (or variant type). To estimate this, it is necessary to know the frequency of the disease in the population, the frequency of the variant (type) in the population, and of course the effect on risk of disease of the variant.

While this consideration is well established for common variants (e.g., ascertained by GWASs), which are seldom used as a clinical test in individual patients, a corollary is that it also applies to rare variants: Just because a variant is rare this does not imply that its presence must have a large effect on disease risk. There is a growing literature reporting the presence of rare, predicted (and sometimes functionally demonstrated) damaging[3] variants ascertained in patients presenting with rare kidney diseases, but without Mendelian transmission of the trait. Such reports (especially where the experimental evidence that the variants affect protein function is strong) may lead clinicians, patients, and health insurers/providers to the conclusion that the identification of such variants is clinically actionable in an individual. However, given the potentially extremely consequential decisions regarding life planning, reproduction, or living donor transplantation that may be made on the basis of assigning such a diagnosis, **outside the setting of Mendelian disease diagnostics, extreme caution is recommended before reporting molecular findings clinically**. This consideration can be illustrated quantitatively by data from large-scale genome sequencing studies in people with urinary stone disease.

COMMON AND RARE NON-MENDELIAN CONTRIBUTORS TO URINARY STONE DISEASE

Urinary (or kidney) stones are a common medical problem with a lifetime prevalence of around 10% in some countries (see Chapter 40 for further discussion on nephrolithiasis).[39] It is known that certain Mendelian disorders (e.g., primary hyperoxaluria, Dent disease, and Bartter syndrome, see later) can cause increased susceptibility to the development of urinary stones, and genetic testing has identified a molecular diagnosis of a Mendelian disorder in up to 10% of patients in some studies, with higher proportion seen in those with younger onset of the condition.[39] It is also known that, even among the 90% or so of sufferers without an explaining monogenic disorder, a family history of stone disease is common (up to 37% and even higher in some studies, compared with <12% of the healthy population) and heritability (i.e., the proportion of disease risk explained by genetic factors) is estimated to be around 45% or higher.[39] Large-scale GWASs (incorporating tens of thousands of subjects) have identified common variants at around 20 different genes that are strongly associated with the condition, but together these variants only explain around 5% of the heritability.[27] Using whole-genome sequencing data from the UK Biobank and the 100,000 Genomes Project, single (monoallelic) truncating or predicted highly damaging rare and ultrarare variants in the gene *SLC34A3* were identified in around 5% of stone formers, compared with approximately 1.5% of controls, with evidence for enrichment of these variants in stone formers highly statistically significant ($P < 4 \times 10^{-10}$).[40] *SLC34A3* encodes the sodium-dependent phosphate transport protein 2C, which is necessary for normal phosphate homeostasis, and biallelic variants in this gene are established as the cause of autosomal recessive hereditary hypophosphatemic rickets with hypercalciuria (HHRH[41]), so the observation of monoallelic rare, damaging variants of this gene predisposing to urinary stone formation is biologically plausible and indeed consistent with previous reports of increased urinary stone disease in parents of affected children with HHRH.[42] However, the presence of such individually rare heterozygous genetic variants also in individuals without urinary stone disease means that identification of such a variant in an individual does not reliably predict future urinary stones, and nor do such variants exhibit Mendelian (i.e., autosomal dominant) transmission of urinary stone disease in families in which they are inherited. They do, however, contribute around 9% of the previously unexplained heritability of kidney stone disease, and the magnitude of the effect of the presence of such a variant on disease risk is similar to (and appears to be additive with) the top decile of polygenic risk score.[40] Such monoallelic rare variants can therefore be usefully considered to be risk factors for urinary stone disease but are not clinically actionable as a monogenic diagnostic finding because they do not have a great enough effect on risk to cause Mendelian inheritance or predict future disease. This is an example where the data indicate that, while clearly having an impact on disease risk, the presence of a monoallelic rare, damaging, or even protein-truncating variant in *SLC34A3* is not a diagnostic finding in the Mendelian sense, and if such a finding is reported to a patient affected by the disease, it is imperative that the magnitude of the risk effect it accounts for is also communicated in a way that the patient understands and that predictive testing in healthy relatives or to allow reproductive interventions is unlikely to be justifiable. It should also be noted that the inference of actionability is critically dependent on the availability of preventive or therapeutic interventions: If in the future a drug or other intervention becomes available that substantially reduces risk specifically in carriers of such monoallelic variants, then the case for predictive genetic testing would need to be reappraised in light of the risks and benefits of the intervention. Monogenic causes of kidney and urinary tract stone disease are considered separately below.

APPROACHES TO GENETIC TESTING IN SPECIFIC PRESENTATIONS

A number of kidney diseases (Table 44.4) including most cystic kidney diseases, ciliopathies, nephronophthisis, tubulopathies, and metabolic diseases have a clear Mendelian basis with a high diagnostic yield on molecular testing. These disorders are discussed in detail in their respective chapters. However, some frequently encountered phenotypes, including familial hematuria, nephrotic syndrome, congenital anomalies, and complementopathies have a more complex range of rare and common genetic variants that can be identified in patients, and clinical approaches to genetic testing in patients with these phenotypes are discussed next.

ALPORT SYNDROME AND FAMILIAL HEMATURIA

Alport syndrome is the second commonest Mendelian cause of kidney failure and is linked to changes in three

[3]A damaging variant is one that affects protein structure or function. This does not necessarily mean the variant is sufficient to cause disease, but it may be used as evidence to support such an assertion.

Table 44.4 Categories of Genetic Kidney Diseases

Category	Example Diseases	Example Mendelian Gene(s)	Notes
Syndromic renal tract developmental disorders (see Chapters 1 and 71)	Townes-Brocks, renal coloboma, renal cysts, and diabetes	*SALL1, PAX2, HNF1B*	AD or AR, highly variable penetrance and expressivity
Nonsyndromic renal tract developmental disorders (see Chapter 71)	Renal hypodysplasia, renal aplasia, posterior urethral valves	*HNF1B, PAX2*	AD or AR, highly variable penetrance and expressivity, rarely Mendelian
Cystic kidney disease (see Chapter 45)	AD polycystic kidney disease	*PKD1, PKD2, IFT140*	AD, penetrance, and severity of PKD1/2 much greater than other causes
	AD polycystic liver disease	*PRCKSH, SEC63, ALG8*	AD, incomplete penetrance
	HNF1B	*HNF1B*	AD, frequently associated with dysplasia/renal asymmetry and extrarenal features
	AR polycystic kidney disease	*PKHD1*	AR, renal cysts sometimes seen in heterozygous carriers
	Kidney cancer syndromes	*VHL, MET, FH, FLCN, SDHB/C/D*	AD, cancer risk, and extrarenal features vary substantially between different disorders
	Tuberous sclerosis complex	*TSC1, TSC2*	AD, renal-limited presentation possible but rare
Ciliopathies (see Chapter 45)	Multisystem ciliopathies	*BBS1, BBS2, ARL6, CEP290, IFT140*	AR
	Isolated nephronophthisis	*NPHP1, NPHP4*	AR
Tubulointerstitial kidney diseases (see Chapter 37)	AD Tubulointerstitial Kidney Disease	*UMOD, MUC1, REN, DNAJB11, HNF1B*	AD, mitochondrial, highly penetrant
Tubulopathies (see Chapter 71)			AD, AR, X-linked, usually highly penetrant with distinctive biochemical or extrarenal features
Metabolic disorders and amyloidosis (see Chapter 35)	Cystinosis, Fabry, primary hyperoxaluria, hereditary amyloidosis	*CTNS, GLA, AGXT, TTR*	Various modes of inheritance and extrarenal features
Inherited glomerulopathies (see Chapters 30, 71)	Alport syndrome	*COL4A3, COL4A4, COL4A5*	X-linked, AR; AD with reduced penetrance for CKD; hematuria, deafness, and EM features
	Podocytopathies	*NPHS1, NPHS1, WT1, INF2, TRPC6 (See Figure 45.6)*	Monogenic causes can occur at any age but particularly enriched age <1 year
Complementopathies (see Chapter 36)	Atypical hemolytic uremic syndrome	*CFH (monoallelic)*	Rare genetic variant detected in 50%-75% of cases, penetrance usually very low, Mendelian inheritance described
	C3 glomerulopathy	*CFHR5, CFH (biallelic), C3*	These Mendelian causes are rare; risk effects of rare and common variants in other genes incompletely understood
Unexplained kidney failure		*INF2, UMOD, CLCN5, CLDN16, PAX2*	15-25% have monogenic cause identified (depending on setting/selection criteria)

AD, Autosomal dominant; *aHUS,* atypical hemolytic uremic syndrome; *AR,* autosomal recessive.

genes: *COL4A3*, *COL4A4* (which are situated adjacent to each other on chromosome 2), and *COL4A5* (which is situated on the X chromosome). Each gene encodes an α chain of type IV collagen (termed collagen α3[IV], α4[IV], and α5[IV,] respectively), which together form the type IV collagen trimer termed collagen α3α4α5(IV), which is first produced in the kidney after birth when podocytes switch from making the collagen α1α1α2(IV) isoform to generate the postnatal glomerular basement membrane. Biallelic (i.e., homozygous or compound heterozygous) pathogenic variants in either *COL4A3* or *COL4A4* (diagnostic of autosomal recessive Alport syndrome, ARAS) or a single hemizygous variant of *COL4A5* (i.e., in a male patient and diagnostic of X-linked Alport syndrome, XLAS) result in failure to produce any normal collagen α3α4α5(IV). This results in hematuria (typically first observed in early childhood) with a high likelihood of subsequent proteinuria, glomerular damage, impaired kidney function, and kidney failure, which is experienced before the age of 30 in more than half of affected individuals.[43] In addition, because collagen α3α4α5(IV) is important in the ear and eye, patients frequently experience hearing loss and some will have eye problems. Indeed, it was the combination of hereditary nephritis and deafness that allowed the definition of Alport syndrome clinically in the early 20th century. Although the distribution across the body of type IV collagen isoforms containing α5(IV) is different from those containing α3(IV) and α4(IV), it is not usually possible without specialized immunostaining of biopsy material to distinguish, in an individual male patient, whether he has the autosomal recessive or X-linked recessive form of the disease (although clearly the family history may reveal this). In addition, as mentioned earlier, females with a single faulty *COL4A5* allele frequently exhibit features of the disease and are at increased risk of kidney failure, which occurs at a median age of around 65 years.[43]

In addition to autosomal recessive and X-linked forms of the disease, it is now clear that heterozygosity for a rare, damaging *COL4A3* or *COL4A4* variant can be associated with a renal phenotype, especially hematuria. Where individuals undergo kidney biopsy, there is almost always thinning of the glomerular basement membrane that is visible on electron microscopy (although not all individuals with such biopsy appearances will have an identifiable change in a *COL4A3/4/5* gene). In addition, some patients will exhibit proteinuria (usually later in life) and if a kidney biopsy is performed at that time, podocyte foot process effacement with or without FSGS may be seen. In some such individuals there will be progressive nephron loss and kidney failure can result. This condition has variously been termed “ARAS carrier state” or “thin basement membrane nephropathy (TBMN)” and sometimes (if there is multigenerational kidney failure in the family) “autosomal dominant Alport syndrome,” although deafness or eye involvement has rarely been reported in such individuals. Previously, the term “benign familial hematuria” was used to describe this condition, but it is now widely recognized that the word “benign” in this context is misleading since a proportion of patients will develop clinically significant kidney disease. The nomenclature for this condition (i.e., the result of heterozygosity for a single *COL4A3* or *COL4A4* pathogenic variant) is still under discussion, but whatever is agreed by the community, clearly the risk of clinically significant disease in such people lies somewhere between that of the general population and that of females with X-linked Alport syndrome and considerably less than ARAS or for males with XLAS. Although initial reports estimated a lifetime risk of kidney failure among *COL4A3* or *COL4A4* heterozygotes as high as 20%,[44] these were based on data from patients who had undergone genetic testing to ascertain their cause of kidney disease, so they were unavoidably enriched for those with a personal or family history of more severe disease course. More recently, analysis of large-scale population sequencing data indicates that up to 0.94% of the general population have a heterozygous variant in *COL4A3* or *COL4A4* that is associated with hematuria,[45] and the lifetime risk of kidney failure in such individuals is now estimated to be below 3%.[46]

The fact that heterozygous variants in the *COL4A3,4,5* genes can be seen in the general population and are associated with a relatively low risk of kidney failure illustrate the difficulties in variant interpretation and highlight again the importance of close collaboration between the clinician and geneticist. If a heterozygous (likely) pathogenic variant is identified in a patient with kidney failure, as occurred, for instance, in a substantial proportion of the “solved” cases in one prominent study of more than 3000 patients with kidney failure,[3] how do we know that such variant is actually causative? Clinical correlation is again essential. In a patient with hematuria, proteinuria, and progressive CKD (i.e., a phenotype consistent with Alport syndrome), the variant is likely causative. But what if the patient had CAKUT with only mild proteinuria? Is the variant causative and the patient have CAKUT and a (clinically inapparent) type IV collagen gene defect? Or is it a modifier? Or just an incidental finding? This example again shows the importance of close collaboration between the clinician and the testing geneticist and that a genetic diagnosis is not a simple black-and-white result that can be done in isolation from the clinical information. For this reason, many centers have regular meetings between clinicians and the genetics laboratory to jointly discuss the relevance of any identified variants in each patient. This is analogous to regular meetings between clinicians and pathologists to discuss the changes seen in the kidney biopsy of a given patient.

For an individual female patient presenting with isolated hematuria and/or a kidney biopsy showing thinning or other abnormalities of the glomerular basement membranes without a genetic test, it is not usually possible to determine whether the cause is a fault in *COL4A5* (that would indicate underlying XLAS) or relates to an autosomal *COL4A3* or *COL4A4* genetic variant, although a careful family history can sometimes be extremely informative. Although the clinical management and indeed prognosis might be broadly similar if the cause were a single heterozygous change in *COL4A4, COL4A4,* or *COL4A5,* if she were to have children, the risk to them would differ substantially depending on the gene: In the presence of an underlying pathogenic *COL4A5* variant, her male offspring would have a 50% chance of inheriting XLAS. This information should be shared with the patient at the appropriate time to inform her decision making. Conversely, in a male patient presenting similarly, distinguishing X-linked from autosomal disease would significantly inform the risk to any (current or future) male and female offspring.

The optimal criteria for offering genetic testing in patients found to have persistent hematuria of glomerular origin remain uncertain. Important indicators of an underlying Mendelian cause include a kidney biopsy showing glomerular basement membranes that are either thin or have abnormalities characteristic of Alport syndrome; a family history of hematuria or unexplained kidney disease; or extrarenal features of Alport syndrome. The choice of whom to offer testing to will depend on the clinical utility in an individual patient and particularly the certainty with which a *COL4A5* variant can be excluded clinically, as well as whether urinalysis of relatives is available. Clearly, given the range of possible outcomes from genetic testing, careful counseling is needed before performing the test and, in patients too young to engage with these issues (and therefore make an informed choice about testing) and no proteinuria, an argument can often be made in favor of deferring testing or, where appropriate, seeking a genetic diagnosis in a relative. If proteinuria has developed at the time of presentation and there are no extrarenal features or affected relatives, a kidney biopsy will usually need to be considered because the differential diagnosis would include IgA nephropathy and other inflammatory glomerulopathies for which specific treatments may be beneficial. Thus for example, it has been observed that common alleles affecting genetic risk of IgA nephropathy are enriched in healthy participants of UK Biobank with isolated hematuria (i.e., with no other clinical features of kidney or structural urinary tract disease) such that the attributable rate of undiagnosed IgA nephropathy may have reached 20% in this study cohort.[47]

While the clinical implications of finding a pathogenic *COL4A5* or biallelic *COL4A3* or *COL4A4* variants in a patient are usually relatively clear, the best advice to offer individuals diagnosed with a single heterozygous *COL4A3* or *COL4A4* variant is less well understood. It seems likely that clinical features in the patient at the time of diagnosis, as well as their own family history, will substantially influence prognostication. It is hoped that better understanding of environmental, phenotypic, and genetic predictors of outcomes in this increasingly common situation will allow this question to be better answered in the future. Alport syndrome is discussed further in Chapter 71 and Box 44.2.

CONGENITAL URINARY TRACT ANOMALIES

Developmental anomalies represent the commonest cause of childhood kidney failure and thus represent a core component of pediatric nephrologic practice, sometimes together grouped into the diagnostic category of Congenital Anomalies of the Kidney and Urinary Tract (CAKUT). Many of these disorders can be associated with extrarenal organ involvement, particularly the lower urinary tract, eye, ear, and neurologic system that can make clinical management (including transplantation) complex. Congenital urinary tract developmental anomalies can occur as part of well-described Mendelian syndromes including urofacial syndrome (associated with biallelic variants of *HPSE2* and *LRIG2*[48,49]), with diabetes and female genital malformations in *HNF1B*-associated kidney disease,[50] and as the renal coloboma syndrome associated with monoallelic variants of *PAX2*.[51] They can also be seen in association with an SV or larger chromosomal abnormality, in which case extrarenal (particularly neurodevelopmental and sometimes cardiac) manifestations are more likely to be present, exemplified by the recurrent 17q12 deletion that includes *HNF1B* and adjacent genes.[52] Mostly, however, developmental renal tract anomalies occur in isolation, in which situation a Mendelian diagnosis is less likely to be detected. Specific genes, phenotypes, and mechanisms associated with congenital renal and urinary tract anomalies are discussed in more detail in Chapter 71.

Congenital urinary tract anomalies frequently cluster in families, and where there is a clear Mendelian pattern of inheritance in a family identification of an underlying molecular diagnosis of a monogenic disorder is more likely. Where there is extrarenal involvement (particularly neurodevelopmental delay) that does not suggest a specific Mendelian syndrome, assays for SVs (including microarrays, karyotyping, or genome sequencing) may detect a complex or multiple gene–involving variant. However, as with the well-defined monogenic syndromes, a renal-limited presentation is possible with larger SVs (e.g., those that involve the *HNF1B* gene).

Several whole-exome and whole-genome sequencing studies in cohorts of patients ascertained for congenital urinary tract malformations have been reported, and estimates of the diagnostic yield of genomic testing range from as high as 25% among patients with more severe disease[53] to 7% with less selection of cases included.[54] Where direct comparisons in diagnostic yield have been made, it is clear that in patients with nonfamilial, nonsyndromic renal tract anomalies, the chance of making a monogenic diagnosis using genomic testing is well below 10%.[55] What is not always clear is what the variant attributable risk of disease is for those variants identified in cases-only sequencing studies for which the frequency of similar variants in the general population is not ascertained. This complicates the assessment of clinical implications when such variants are identified in an individual or family. When considering prenatal counseling, because of the wide spectrum of phenotypes and severity associated with monogenic forms of CAKUT (even within the same family), a genetic diagnosis cannot always reliably predict future (especially extrarenal) organ involvement, so clinical monitoring, for instance using ultrasound scanning, can be of greater value to predict outcomes during and shortly after gestation. However, for some monogenic syndromes, particularly those associated with the most severe manifestations, such as renal tubular dysgenesis or bilateral renal aplasia, a genetic diagnosis can help with decision making and enable preimplantation genetic diagnosis to be offered for subsequent pregnancies.

In addition to identification of rare variants affecting renal tract development, across numerous published series it is clear that unbiased genomic testing can sometimes identify other, unexpected and clinically actionable monogenic disorders (including nephronophthisis and polycystic kidney diseases) among patients ascertained as having congenital or developmental renal anomalies.[55,56] Given this, while there is mounting evidence of clinical utility of genomic testing in patients presenting with congenital urinary tract anomalies, the precise clinical criteria; the tests to run (panel, CNV microarray, whole-exome or whole-genome sequencing); and, critically, how each variant identified should be interpreted, are still areas of uncertainty. At the current time, the clinical value of molecular testing for congenital urinary tract anomalies requires careful consideration for each patient on an individual basis, considering their and their family's clinical features, the molecular analysis methodologies available, and the ability to interpret the actionability of genetic variant(s) identified. It is hoped that future large-scale genomic analysis efforts will address some of the knowledge gaps in this area.

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE

Autosomal dominant tubulointerstitial kidney disease (ADTKD) is notable because it can frequently manifest

Box 44.2 Clinical Assessment and Management of Patients With Alport Syndrome

Alport syndrome is the second commonest Mendelian cause of kidney failure (after autosomal dominant polycystic kidney disease) and is associated with kidney failure and other problems as early as childhood in some patients. As ever-larger series and registries of patients with Alport syndrome have been published, it has become clear that there are strong genotype-phenotype associations in the disease, with the type of mutation (in addition to the zygosity) being strongly associated with outcomes.[153] Among patients with autosomal recessive Alport syndrome (ARAS) or males with X-linked Alport syndrome (XLAS), the commonest type of missense mutations are substitution of glycine (which is the smallest amino acid with a side chain composed of a single hydrogen atom) with a bulkier residue. Glycine residues account for more than one-third of the amino acids in collagenous domains of each collagen protein chain and are necessary to allow tight winding of the three chains to form the triple helical collagen α3α4α5(IV) protein, so glycine substitutions result in distortion of this shape, disrupting function and leading to clinical manifestations. However, since the collagen subunits trimerize via their C-terminal noncollagenous (NC) 1 domain, variants that truncate the peptide result in a much stronger loss-of-function effect since they prevent incorporation of the entire polypeptide chain into the trimer. Clinically, such variants (and indeed some NC1 domain missense variants that have a large effect on protein structure, impairing trimer-formation) are associated with kidney failure at an earlier age than collagenous domain glycine substitutions. However, there is still significant spread and overlap in terms of age at kidney failure between these categories, so while useful as a guide to risk, care must be taken when advising patients that disease course in an individual may be very different from the average for a particular type of genetic variant. Among all truncating mutations, nonsense (as opposed to frame shifting, deletion, or splicing) mutations appear to have the worst prognosis.[154] It is not completely understood why this is the case.

While progressive loss of renal function over time allows patients with Alport syndrome to be classified as having chronic kidney disease (CKD), there are important differences between the populations of people with Alport syndrome compared with the general (i.e., unselected) CKD population: First, patients with Alport syndrome tend to be younger (at each level of kidney function); second, renal function (even stratified for age) tends to decline much more rapidly in patients with Alport syndrome, with accelerating nephron loss especially evident when estimated glomerular filtration rate (eGFR) drops below around 30 mL/min; third, in patients with ARAS and males with XLAS, proteinuria tends to increase over time, even before eGFR begins to drop. Together, these factors contribute to a much higher risk of future kidney failure at any given level of renal impairment among people diagnosed with Alport syndrome compared with unselected CKD cohorts. This massively increased risk of kidney failure indicates significant unmet need for effective therapies that protect the kidney in Alport syndrome and also justifies a more aggressive approach to available renal protective therapies, especially early in the disease, than might be considered proportionate in the setting of more common causes of CKD.

It is becoming clear that heterozygous individuals are also at increased risk of kidney disease, but the magnitude of this risk is still uncertain: Systemic review of the case series suggests that females with XLAS have an approximately 15% lifetime risk of kidney failure (at median age of 65 years), although this methodology is susceptible to the effect of ascertainment bias.[155] It is known that women frequently exhibit worse proteinuria and sometimes decline in renal function during pregnancy, presumably owing at least in part to the effect on their susceptible filtration barrier of the associated hyperdynamic state. Patients heterozygous for pathogenic variants in *COL4A3* or *COL4A4* often exhibit microscopic hematuria (with thinning of GBMs almost always present on electron microscopic examination of a kidney biopsy, with thickening/lamellations only seen occasionally and segmentally in some patients) and are known to be at increased risk of proteinuria (in which situation a kidney biopsy can often show secondary FSGS, usually with segmental foot process effacement) and even subsequent kidney failure. Although numerous case series exist showing this association[156] and there is relatively high occurrence of heterozygous *COL4A3* or *COL4A4* variants detected in adults presenting with unexplained kidney failure,[3] unlike hematuria, kidney failure rarely shows Mendelian (i.e., autosomal dominant) inheritance within families of such individuals. Large-scale genomic sequencing datasets from general populations reveal that almost 1% of some populations are heterozygous for pathogenic variants in *COL4A3* or *COL4A4,*[45] suggesting the lifetime risk of kidney disease is probably below 3%[46] (contrasting with figures in the older literature as high as 20%[157] that likely resulted from ascertainment bias owing to individuals with a personal or family history of significant kidney disease being more likely to have undergone genetic testing). Because of this complexity, there is currently controversy around the nomenclature of the disease associated with autosomal heterozygosity for variants of *COL4A3* or *COL4A4*. The historical term "benign familial hematuria" is now thought likely to be misleading since there clearly is an increased risk of kidney failure, and the older proposal to use the label "autosomal dominant Alport syndrome" is becoming less well justified as appreciation of the low absolute risk (with lack of Mendelian inheritance) of clinically significant kidney disease (or indeed deafness, see below) in most families together with the consideration that a diagnostic label of "Alport syndrome" will likely preclude life insurance, even if the actual risk of kidney failure in most individuals with such genetic variants is now known to be extremely low. An alternative term that has been used for many years is "thin basement membrane nephropathy," which, although some have argued is only justified in those individuals who have undergone a kidney biopsy, has the advantage of being applicable to patients with relevant clinical and biopsy features in whom genetic testing is either not available or does not reveal a (likely) pathogenic change in a relevant gene. Clearly, it is important to recognize that the term "thin basement membrane nephropathy" should not be inferred to automatically imply a benign prognosis, and long-term surveillance (to detect any hypertension and proteinuria, at least into late middle-age) is indicated in any individual with this condition.

Clinical trials using the angiotensin-converting enzyme inhibitor ramipril have provided evidence that early institution of renin angiotensin system blockade can delay kidney failure in males with XLAS.[158] This recapitulates evidence from mouse models[159] and fits with a paradigm in which increasing pressure across the glomerular filtration barrier accelerates nephron loss in the disease owing to the defective extracellular matrix composition in the GBM, so it is unsurprising that targeting this process pharmacologically provides clinical benefit. Clinically, the data have been interpreted by professional organizations (such as the European Renal Association), leading to recommendations that all patients with any evidence of microalbuminuria or proteinuria in the setting of X-linked or autosomal recessive Alport syndrome, or a single heterozygous *COL4A3* or *COL4A4* variant, should be offered this treatment. In addition, newer guidelines have advocated institution of renin angiotensin system blockade from the age of 2 years if hematuria is evident, even in the absence of any proteinuria, in hemizygous males with XLAS or any patient with ARAS.[160] There is currently less strong evidence for the use of sodium glucose

Box 44.2 Clinical Assessment and Management of Patients With Alport Syndrome (Continued)

transporter 2 inhibition in Alport syndrome, but the magnitude of the benefit in terms of reduction in progression to kidney failure among adult patients with proteinuric chronic kidney disease (and the fact that patients with Alport syndrome were not excluded from large-scale trials of this therapy[161,162]) has provided justification for its use in clinical practice, with a retrospective case series suggesting that it is well tolerated in this group.[163] It is hoped that trials of this class of medication in children with kidney disease will provide direct evidence that allows quantification of safety and efficacy in Alport syndrome.

As a multisystem disorder affecting hearing and the eyes, as well as the kidney, XLAS and ARAS can have significant implications for affected children and young people and guidelines recommend annual childhood screening for hearing impairment from the age of 4 years in boys diagnosed with XLAS and all children with ARAS, since language and educational development can be impacted if hearing loss is unrecognized in childhood. Frequent surveillance for visual impairment is not mandated given existing childhood screening programs for myopia in many countries,[160] but an awareness of the risk of keratoconus, which may manifest as severe myopia, is important for patients and parents of affected children. Eye and ear complications are rare in people with a single autosomal pathogenic variant, and although hearing loss is commonly seen in older individuals, alternative causes (genetic or acquired) should be sought in young people in this category with significant hearing impairment.

If kidney failure occurs, the best current treatment modality is kidney transplantation, but since many patients will have affected (or heterozygous) first-degree relatives, access to living-related kidney donors is reduced among people with Alport syndrome compared with non-Mendelian causes of kidney disease. An important question relates to the suitability of heterozygous relatives (for instance, mothers of boys with XLAS) to donate a kidney. Guidelines suggest that other options should be explored first, and the pros and cons of predonation kidney biopsy should be discussed with such donors in order to identify subclinical evidence of kidney damage. While there is no empiric evidence that allows quantification of the effect on risk of future kidney disease in heterozygotes donating a kidney, it is generally assumed that it is greater than for people of similar health, age, and sex with normal type IV collagen genes, so decisions should be made taking into account both this unquantified risk and also the potential benefit to the recipient. Data revealing whether allografts from heterozygotes fare less well are also lacking, further affecting the quality of advice that can be shared with patients in these discussions. Having said that, kidney donation by heterozygous females who might become pregnant in the future is generally discouraged since the combined risk factors of their genotype and having a single kidney during pregnancy are difficult to quantify.

A rare complication of kidney transplantation in patients with Alport syndrome is *de novo* production of an antibody response directed against type IV collagen of the allograft glomerular basement membrane. With current immunosuppression regimens, this risk is now estimated to be below 3% and far lower in patients with at least one missense variant.[160] While immunohistochemistry can show linear staining of the glomerular basement membranes, serologic tests for antiglomerular basement membrane antibodies are not thought to be sensitive in this situation unless the biallelic missing chain is collagen a3(IV), which contains the Goodpasture antigen, which is detected serologically.

without any distinctive phenotypic features: Most patients exhibit progressive reduction in glomerular filtration rate (GFR), the onset of which is typically in early to mid-adulthood, with kidney failure usually occurring in midlate adulthood, typically before the age of 60 years. Clinical presentation of ADTKD is usually with impaired GFR in the absence of urinary abnormalities (i.e., without hematuria or proteinuria) and more often than not with normal blood pressure (although this may rise if not treated as affected individuals approach kidney failure). Kidney biopsy almost always shows tubular atrophy and interstitial fibrosis (often in proportion to the degree of renal impairment) but no evidence of glomerulopathy, beyond globally sclerosed glomeruli as nephrons are lost.

The commonest causes of ADTKD are missense variants in the *UMOD* gene and variants that introduce a shift in the reading frame of a variable number tandem repeat within the *MUC1* gene. ADTKD-*UMOD* is associated in an appreciable number of, but by no means all, patients with gout early in life—even when GFR is not significantly reduced, which distinguishes this condition from almost all other causes of CKD, in which gout is common only after GFR is substantially reduced. In addition to gout, some individuals will exhibit kidney cysts, most often after GFR is substantially reduced and without renal enlargement. Previous names for this condition include “familial juvenile hyperuricemic nephropathy” and “medullary cystic kidney disease type 2,” but both terms have been superseded by ADTKD-*UMOD*.[57]

ADTKD-*UMOD* can only be reliably diagnosed in clinical practice by identification of a pathogenic or likely pathogenic change in the *UMOD* gene, almost all of which are in exons 3 and 4 of the gene and which are most often a missense variant that results in loss (or sometimes gain) of a cysteine residue. They likely mediate disease via a gain-of-function mechanism by preventing normal folding of the uromodulin protein in the endoplasmic reticulum of the renal tubular cells, where it is produced in abundance, leading to activation of the unfolded protein response and, eventually, cell death.[58] Of note, population genetic studies have shown that uncommon and common genetic variants of this gene are also associated with altered risk for CKD without the phenotypic features or monogenic inheritance pattern of ADTKD.[59-61]

Linkage of ADTKD (at the time termed medullary cystic kidney disease type 1) to a locus on chromosome 1q21 was reported in 1998,[62] but it took until 2013 before the genetic changes that cause this condition were first identified in the *MUC1* gene.[63] The reason for this extended delay was the technical difficulty posed by detection of the responsible variants in the gene. *MUC1* encodes mucin-1, which contains a heavily glycosylated extracellular repeat domain encoded by a 60 base-pair variable number tandem repeat (VNTR), with most humans having between 25 and 125 copies of on each allele. This VNTR has a guanosine and cytosine (which form three hydrogen bonds when base-pairing) content of more than 80%, and the presence of so many strong bonds results in secondary structure of the DNA strands that makes amplifying and sequencing this exon challenging and is not usually possible using PCR or short-read massively parallel sequencing approaches. The initial *MUC1* mutations identified in families with ADTKD result in a single cytosine insertion following a 7-cytosine homopolymer within one copy of the repeat, introducing a shift in the reading frame of the protein that results in production of a neopeptide believed to cause disease via a gain-of-function mechanism by activating the unfolded protein response that is toxic to tubular cells.[64] The first available assay to detect *MUC1* VNTR frameshifting variants was specific for cytosine insertions, but more recently multiple other variants that all result in production of a similar frameshifted peptide have been reported. These can be detected by immunostains specific to the pathogenic peptide of urinary cells[65] or using long-read sequencing approaches such as single-molecule real-time sequencing.[66] Availability of such assays for clinical diagnosis remains limited, but hopefully it will become better in the future. Critically, short-read sequencing approaches (including gene panels, whole exomes, or whole genomes) have extremely limited sensitivity for the diagnosis of ADTKD-*MUC1*. Because ADTKD-*MUC1* is not associated with phenotypic abnormalities in childhood and in the absence of a proven preventative intervention, testing of children unable to provide informed consent for predictive testing is not usually recommended.

Genetic variants in the *REN* gene that encodes renin and its prohormones can also present with ADTKD, although this is much less common than *UMOD* or *MUC1*-associated disease. Patients with ADTKD-*REN* can present in childhood with low or normal blood pressure and failure to thrive, sometimes with associated hypokalemia and anemia with low renin levels on biochemical testing. There is increased susceptibility to acute kidney injury in the setting of hypovolemia or intercurrent infection in early life. Reduced eGFR is frequently observed, and where the mutation is in the signal peptide or preprorenin propeptide, this can be as early as infancy, although even in this group kidney failure is rare before the age of 35 years, with later onset and slower progression seen in individuals with mutations in the parts of the gene encoding prorenin or the mature renin hormone. Aberrant processing of the mutant peptide can result in its intracellular deposition, leading to cellular stress and apoptosis of the juxtaglomerular apparatus cells (suggesting a gain-of-function mechanism) leading to significantly decreased plasma renin activity and CKD.[58]

Other genes reported in association with ADTKD include *HNF1B* and *DNAJB11,* which had both previously been reported in cystic kidney disease. In addition, tubulointerstitial kidney disease can be the presenting or salient feature of disease associated with further genes (such as *SEC61A1*, *JAG1*, *NOTCH2,* or *SALL1*) that are usually associated with distinctive extrarenal manifestations. However, since the phenotype of these disorders can overlap with presentations of ADTKD, they remain in the differential diagnosis in families presenting with ADTKD.

Families with homoplasmic or near-homoplastic mitochondrial DNA mutations that affect mitochondrial tRNA (particularly for leucine or phenylalanine) synthesis can also present with tubulointerstitial kidney disease and familial kidney failure, termed mitochondrially inherited tubulointerstitial kidney disease (MITKD), because of the pattern of inheritance they exhibit.[67] Usually, such patients do not exhibit renal Fanconi syndrome, although electrolyte abnormalities have been reported in some families including a Gitelman-like syndrome.[68]

Because of the lack of specific phenotypic abnormalities, genetic testing in ADTKD is especially useful, especially for predictive testing, and should be considered where there is autosomal dominant inheritance of nonproteinuric, progressive kidney disease (and similarly MITKD where the pattern of inheritance suggests a mitochondrially encoded disorder). However, perhaps because the combination of impaired kidney function with no urinary abnormalities is commonly seen in the general population (e.g., due to obstructive uropathy, prior episodes of acute kidney injuries, hypertensive nephrosclerosis, or renovascular disease), genetic testing for ADTKD in individuals who lack a family history generally has a low likelihood of providing a diagnostic finding and so, for reasons discussed earlier, is usually avoided. Clinical features of tubulointerstitial diseases are discussed in greater detail in Chapter 37 and cystic diseases are discussed further in Chapter 45.

ATYPICAL HEMOLYTIC UREMIC SYNDROME

Linkage of familial atypical hemolytic uremic syndrome to the *complement factor H (CFH)* gene in the 1990s[69] provided key mechanistic insight into the pathophysiology of this disorder, contributing significantly to the development of therapeutic terminal complement pathway blockade that has shown striking evidence of efficacy in this condition (aHUS, discussed in detail in Chapter 36).[70] Since that initial family study, other families with loss-of-function *CFH* variants cosegregating with aHUS have been reported, as have rare loss-of-function variants in *CD46* (which encodes membrane cofactor protein[71]), and rare gain-of-function *complement factor B* (*CFB*) and *C3* variants.[72,73] In addition, SVs that result in replacement of the C-terminal domains of factor H with the homologous domains of factor H–related protein (FHR) 1 or FHR3 (encoded by the *CFHR1* and *CFHR3* genes, respectively)[74,75] are also reported in familial aHUS, although in many of the reported families, penetrance was incomplete. Rare, monoallelic loss-of-function variants of *complement factor I (CFI)* have also been reported in cohorts of patients with aHUS,[76] but more recent analyses of large biobanks suggest that the penetrance of such genetic defects is very low indeed[77] and these defects are rarely, if ever, reported to show a Mendelian pattern of inheritance of aHUS. In addition to disorders of genes controlling the complement system (where there is usually a good response to therapeutic terminal complement pathway blockade), biallelic rare loss-of-function variants of *DGKE*[78] and *MMACHC*[79] have been reported in childhood-onset aHUS (especially below the age of 1 year) in which response to terminal pathway blockade is limited. Rare variants in thrombomodulin (THBD), *CFHR5,* and plasminogen (PLG) have also been reported in cohorts of patients with aHUS, but the evidence for Mendelian transmission of aHUS in these cases is not as well established, and when compared with the frequency of rare variants in these genes in the general population, evidence of enrichment has not been reported,[80] perhaps explained by lower absolute risk of aHUS conferred by variation in these genes.

Polymorphic deletions of *CFHR1*, *CFHR3*, and *CFHR4* are common in the general population (present in around 18% of UK and 54% of Nigerian individuals[81]), and GWASs with subsequent fine mapping studies showing this allele are associated with reduced risk of IgA nephropathy.[82] Interestingly, biallelic lack of *CFHR1* is associated with increased risk of developing autoantibodies against factor H that are themselves a cause of aHUS.[83,84] The mechanisms underlying these observations are not completely understood, but these relationships with disease illustrate the potential complexity of genetic perturbations of the complement system in humans.

Because the complement system is in a state of finely balanced equilibrium between activation and regulation, it is thought likely that common variation at the genetic loci encoding its key constituents (particularly *CFH, CFHR1-5, CD46,* and *C3,* sometimes termed the "complotype") plays an important role in disease pathogenesis, with observational data in relatives of affected index cases in one cohort (of whom 10 out of 134 sharing a proven or likely functional rare variant exhibited disease, consistent with penetrance of these variants of well below 10%) supporting this hypothesis and suggesting an additive effect on aHUS risk of rare and common variants at the relevant loci.[85] However, currently there is a lack of population-based case-control data that would allow quantification of the risk effects of the relevant common haplotypes and integration of these effects with those of different types of rare genetic variants, which hampers incorporation of these observations into clinical decision making.

Currently, information from genetic analysis in patients with aHUS can inform clinical practice in several ways including:

1. Guiding potential therapy: Terminal pathway blockade is less effective in *DGKE*- and *MMACHC*-associated disease, and risk of relapse after cessation of terminal pathway blockade is higher in the presence of rare, functionally important *CFH* variants;
2. Prognosis: Recurrence risk after transplantation is higher in the presence of rare *CFH* variants and lower where a likely pathogenic *CD46* variant (or no rare variant) has been identified;
3. Transplantation decisions: aHUS poses a particular challenge when considering living-related kidney transplantation because of reports of *de novo* aHUS in four kidney donors, one of whom shared a rare *CFH* variant with the recipient,[86] and at the time of writing the consensus is to avoid kidney donation by individuals sharing a rare variant risk factor with an affected relative.[87]

In addition, identification of a rare genetic variant that can be established as a strong risk factor for the disease can provide greater confidence where there is uncertainty about the diagnosis. However, despite these areas of important utility, it should be recognized that, because of the low penetrance of most rare variants reported in aHUS, use of molecular testing for predictive diagnosis and reproductive interventions is not usually undertaken outside the situation where there is clear Mendelian pattern of disease within a family.

IMMUNE COMPLEX MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS AND C3 GLOMERULOPATHY

Immune complex membranoproliferative glomerulonephritis (IC-MPGN) and C3 glomerulopathy (C3G) are ultrarare glomerular disorders characterized by glomerular inflammation with deposition of immunoproteins: immunoglobulins and complement C3 in IC-MPGN and C3 with scanty or no staining for immunoglobulins in C3G and are discussed in detail in Chapters 36 and 71. Genetic studies in the extremely rare situation of familial C3G have identified three different types of Mendelian disorders:

1. Dominantly inherited SVs affecting one or more of the five *complement factor H–related (CFHR)* genes that result in an elongated, novel gene either by internal duplication of one of the *CFHR* genes or by generation of a hybrid gene. Examples include duplication of exons 2 and 3 of *CFHR5* (which is endemic in Cypriots, having been reported cosegregating with disease in dozens of families[88,89]); and *CFHR2/5* and *CFHR3/1,* etc. hybrid genes, each reported in a single family.[90-92] In all these cases, the SV results in production of a novel mutant protein, detectable in the blood, that contains a duplication of the N-terminal two domains of the protein, which normally mediate dimerization. Their duplication confers on the mutant proteins the capability of forming trimers or higher-order oligomers that are more potent competitive antagonists of factor H, a key regulator of complement C3, at surfaces.[93,94] The mutations therefore have a gain-of-function effect and exhibit a high likelihood of causing glomerular inflammation, with penetrance of hematuria >90% among the Cypriot population[89];
2. Activating mutations of *C3,* such as c.2768_2773delACGGTG, p.(Asp923_Gly924del) and c.2327T>C, p.(Ile756Thr), have each been reported in a single family, alongside evidence that the mutation prevents normal regulation of the active C3b protein, leading to glomerular inflammation, again by a gain-of-function mechanism[95,96];
3. Recessively inherited (i.e., biallelic) *CFH* variants that result in severe deficiency of factor H have been described with C3G presenting early in life.[97,98] C3G is also observed following experimental manipulations that result in *Cfh* deficiency in mice.[99]

In addition to these Mendelian disorders of complement alternative pathway regulation, monoallelic rare variants in these and several other genes (including *CFH, CFB, CFI, CD46*) have been reported in up to 20% of individuals in published series of (overwhelmingly nonfamilial) C3G,[100-102] although such variants were not shown to be enriched compared with unaffected individuals in one large series.[103] While experimental evidence suggests that some of these rare variants have a measurable effect on complement pathway activation in laboratory systems, and some have also been reported in patients with aHUS, the cumulative frequency of variants in the general population (as represented in GnomAD [104] for instance) that have been reported by laboratories as pathogenic[80] is >1:700, implying that, given the rarity of C3G, on average these variants each convey an absolute risk of disease well below 1%. It is possible that within this group there are some variants that have a greater effect on risk (and therefore some an even smaller effect), but clearly the implications of finding such a variant in an individual patient with C3G are different from making a diagnosis of a Mendelian disorder. In this context, it is advisable for **extreme care** to be taken to ensure that the interpretation of such a finding is in accordance, quantitatively, with all the available evidence. For example, the presence of a monoallelic genetic variant that has a weak or negligible effect (in terms of absolute risk of future C3G) in a child with C3G, as well as their healthy parent, should not, by itself, preclude living kidney donation by that parent if the child reaches kidney failure: consideration in this situation needs to be given to any other shared (common or rare) genetic factors in the family.

MONOGENIC GLOMERULOPATHIES

Since the identification of *NPHS1* (encoding Nephrin) in Finnish children with congenital nephrotic syndrome,[105] monogenic disorders affecting essential components of podocytes are well recognized to cause congenital and childhood-onset nephrotic syndrome. As would be predicted, these conditions do not typically respond to treatment with corticosteroids or other immunomodulatory agents and in the past 3 decades, it is clear that defects in these genes account for an appreciable proportion of steroid-resistant nephrotic syndrome in childhood. The genetic defects identified include autosomal recessive and autosomal dominant conditions, and as more have been identified, it is clear that many monogenic podocytopathies, while causing proteinuria, are not associated with nephrotic syndrome and that many of these disorders are associated with extrarenal manifestations, as summarized in Fig. 44.5. Some of the autosomal dominant disorders typically present in adulthood, and even some autosomal recessive disorders (such as those associated with biallelic *NPHS2* [Podocin] mutations) can present in adolescence or adulthood.

Among published series of patients presenting with steroid-resistant nephrotic syndrome, estimates of the proportion explained by monogenic defects range from 10%[106] to as high as 24%,[107] with the differences potentially explained by whether patients presenting in the first year of life (among whom a monogenic defect is more likely) were included and the degree of enrichment with other features (such as family history and consanguinity) suggestive of monogenic disease. Across all large series to date, it is clear that the detection of a monogenic disorder is strongly predictive of greater likelihood of progression to kidney failure, lack of response to immunosuppressive or corticosteroid therapy, and low risk of recurrence following transplantation, with older reports in the literature of steroid responsiveness almost all being attributable to mistaken attribution of causality of variants subsequently shown to be common in a healthy population.[108] This illustrates firstly

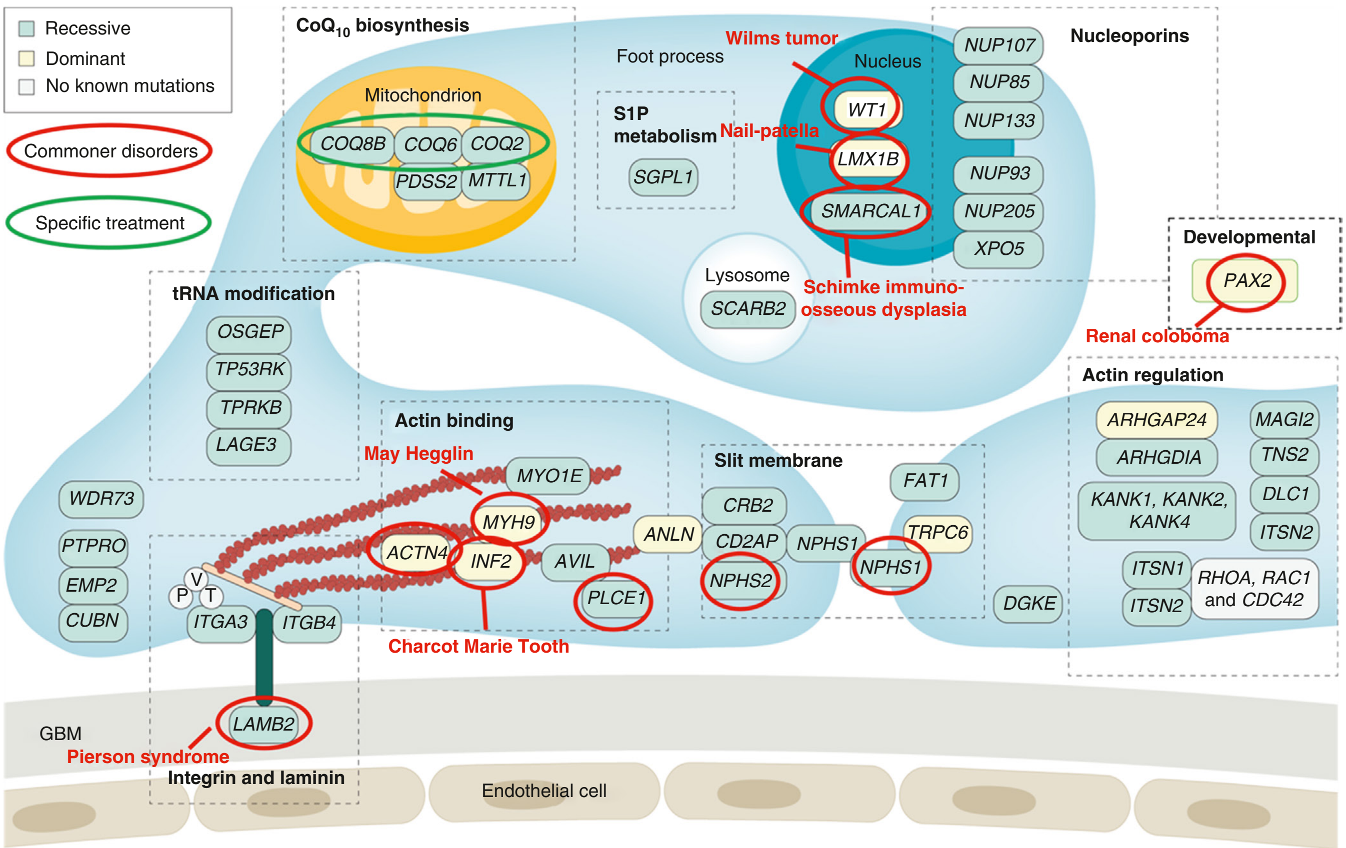

Fig. 44.5 Monogenic causes of podocytopathy. (Adapted from Kopp JB, Anders HJ, Susztak K, et al. Podocytopathies. *Nat Rev Dis Primers*. 2020;6(1):68.)

the value of making a genetic diagnosis (avoiding potentially harmful immunosuppressive therapies and influencing transplantation decisions) but secondly provides a further illustration that not all the variants reported in the literature necessarily fulfill currently recognized criteria for clinical actionability.

The large proportion (75%–90%) of individuals with childhood-onset steroid-resistant nephrotic syndrome (and indeed adult-onset steroid unresponsive primary FSGS) in whom a molecular diagnosis is not detected on genomic testing could theoretically be attributable to undetected or unrecognized Mendelian diseases in this group of patients (e.g., variants in the known disease-linked genes not detected by the analysis strategy used or else variants in genes not currently known to be associated with proteinuric kidney disease). However, evidence from two independent sources argues against most of this unexplained disease being attributable to undetected monogenic disorders. The first is that genetic risk factors for childhood steroid-sensitive nephrotic syndrome (detected by GWAS in this condition and largely driven by strong associations at the HLA locus), while not enriched in patients with identified monogenic glomerulopathies, are similarly enriched in cohorts of patients with 1. childhood-onset steroid-sensitive nephrotic syndrome, 2. adult-onset minimal change disease, 3. adult-onset primary FSGS, and 4. gene test negative for childhood-onset steroid-resistant nephrotic syndrome. This suggests that shared (likely autoimmune) mechanism(s) may be operating in all of these presentations of proteinuric kidney disease.[109] The second line of evidence relates to the recent description of antinephrin autoantibodies detected in the sera of significant proportions of patients with childhood- or adult-onset minimal change disease and adult-onset primary FSGS, including an especially high proportion of those with recurrent FSGS after transplantation.[110,111] Empiric evidence regarding the frequency of antinephrin autoimmunity in genetic test negative for childhood-onset steroid-resistant nephrotic syndrome is eagerly awaited as this may guide future genetic testing strategies in clinical practice. These disorders are discussed in more detail in Chapters 4, 30, and 71.

APOL1-MEDIATED KIDNEY DISEASE

Apolipoprotein L1 (APOL1) is a protein that helps protect against infections by protozoa of the genus *Trypanosoma,* which are responsible for trypanosomiasis, or African sleeping sickness. The protein is thought to act by forming a cytotoxic cation channel in the parasite plasma membrane.[112] There are two key sets of *APOL1* gene variants, designated G1 and G2, which in homozygosity or compound heterozygosity are associated with an increased risk of proteinuric kidney disease. G1 comprises two missense variants (p.S342G and p.I384M) occurring on the same allele, while G2 is defined by deletion of two consecutive amino acids (p.N388del and p.Y389del). Alleles that lack these variants are designated G0. The G1 and G2 variants are only present in individuals with recent African ancestry and confer protection against a broader range of *Trypanosoma* species than the commoner G0 alleles. The heterozygote advantage hypothesis suggests that these variants became common in sub-Saharan populations as a defense against trypanosomiasis, but their effect in homozygosity or compound heterozygosity may contribute to the higher incidence of CKD seen in individuals with recent sub-Saharan African ancestry.

Biallelic G1 or G2 *APOL1* variants are associated with increased risk of a range of renal phenotypes, especially FSGS, occurring in the context of hypertension, HIV infection (HIV-associated nephropathy), and lupus nephritis. For instance, the presence of two *APOL1* risk variants can increase the risk of FSGS by 10 to 17 times and the risk of HIV-associated nephropathy by 29 to 89 times. Although there are suggestions from some studies that even a single G1 or G2 allele might slightly increase the risk of kidney disease, these observations may be explained by confounding (other genetic risk factors for kidney disease are known to be more common in some populations in which *APOL1* risk alleles are common) and the magnitude of any heterozygous effect is smaller and less likely to be clinically actionable.

It is thought that the *APOL1* G1 and G2 variants cause disease owing to gain-of-function mechanisms. While the precise cellular mechanisms are not fully understood, new treatments for *APOL1*-associated FSGS, such as the APOL1 channel inhibitor inaxaplin, suggest that podocytes might be the primary targets in this setting[113] with the abnormal activity of cation channels in the podocyte cell membrane thought to be an initial step in the injury process, followed by inflammation and further damage. Additionally, a protective (i.e., loss-of-function) substitution variant, p.N264K, is present in around 4% of African Americans with a G2-containing risk genotype, and this appears to abrogate the risk effect of G2,[114,115] adding complexity to the assessment of genetic risk.

Clinically, *APOL1* kidney disease presents an unusual situation because, even in the presence of a high-risk genotype (defined as biallelic G1 or G2 alleles in the absence of any p.N264K alleles), the absolute risk of clinically significant kidney disease is small: Biallelic G1 or G2 alleles are present in around 13% of African Americans and more than 80 million people globally, but only a small proportion of these individuals will develop significant kidney disease, and these only in certain settings (including hypertension and HIV infection), so despite the large relative risks associated with high-risk genotypes, the absolute risk of kidney disease associated with *APOL1* is small.

While the presence of a high-risk *APOL1* genotype (i.e., G1:G1; G2:G2; or G1:G2 in the absence of p.N264K) in an individual with kidney disease may provide some explanation for his or her disease, the implications of such a finding are not the same as identification of a Mendelian disease. Specifically, the implications for healthy relatives of such an individual are not completely understood, so this genetic finding must be interpreted in the context of other risk factors for kidney disease (including family history, HIV infection, or other factors) present in individuals. Currently, there remains uncertainty about the place of diagnostic testing in clinical practice. A further area of incomplete understanding is assessment of potential living related donors who may have a high-risk *APOL1* genotype. While absence of this risk factor may provide reassurance (especially in the assessment of individuals with recent African ancestry who have a strong family history of kidney disease in relatives with high-risk *APOL1* genotypes) the possibility that other genetic and environmental risk factors might be present should be considered when discussing the risks of living kidney donation. Conversely, while the presence of an *APOL1* risk genotype in a prospective donor almost certainly implies increased risk of kidney disease in him or her in the future, the magnitude of any increase in risk and the effect of donating a kidney are not currently well understood (risks associated with kidney donation are discussed further in Chapter 70). It is hoped that clinical studies such as the APOLLO study[116] will provide more information in the future to inform decision making in this area. *APOL1*-mediated kidney disease is discussed further in Chapter 75.

TUBULOPATHIES (INCLUDING METABOLIC DISORDERS AFFECTING TUBULAR FUNCTION)

Tubulopathies are typically rare or ultrarare disorders with a highly specific phenotype that is usually linked to a single or only a few genes. While some tubulopathies can be acquired, this is typically later in life and in a recognized risk setting. For instance, nephrogenic diabetes insipidus (NDI, also called AVP-resistance) can be the consequence of long-term lithium intake or distal renal tubular acidosis can develop in the context of Sjögren syndrome, presumably due to autoantibodies against the relevant transporters.[117,118] Yet in the absence of such risk factors, and since most inherited tubulopathies already manifest in early childhood, the *a priori* likelihood that a rare variant or variants identified in a suspected underlying disease gene in an affected child is causative is extremely high. The ClinGen consortium used the example of cystinosis to demonstrate the implications of this for variant interpretation[119]: Cystinosis is a disorder with a typical phenotype and a highly specific diagnostic test (white cell cystine) and linked to a single gene *(CTNS),* in which causative variants are identified in >95% of patients. Thus any two rare variants identified *in trans* in an affected individual have an *a priori* likelihood of being causative of close to 100%. Other such examples would be EAST syndrome or Proximal Tubular Acidosis-Ocular Abnormalities syndrome: Both have highly specific phenotypes that are linked to only a single gene (*KCNJ10* and *SLC4A4*, respectively), so any two rare variants identified *in trans* can be considered pathogenic. For most other tubulopathies, two or more genes can be responsible, although clinical differences in the phenotype can usually establish the most likely underlying gene.[120] The presence of congenital deafness in a child with distal renal tubular acidosis, for instance, would make *ATP6V1B1* the most likely underlying gene, and the history of prematurity and presence of typical but rather mild electrolyte abnormalities in a child with a clinical diagnosis of Bartter syndrome would favor *KCNJ1*.[121,122]

Consequently, the diagnostic yield in tubulopathies is high, exceeding 60% when testing affected children, with some disorders (like cystinosis) approaching 100%, but this drops to around 30% in patients manifesting in adulthood.[123,124] There is also strong agreement between clinical and genetic diagnosis, so genetic testing in most cases simply confirms the clinical diagnosis. Nevertheless, testing is still of benefit, as for many clinicians the patient under investigation may be the first and only patient with this diagnosis they may ever see, given the rarity of most of these disorders; therefore genetic confirmation may provide welcome reassurance. Moreover, the benefits discussed at the beginning of the chapter, such as facilitating informed choices regarding reproductive planning or presymptomatic diagnosis in a subsequent sibling, also apply because there sometimes can be divergence between clinical and genetic diagnosis. For instance, Gitelman syndrome, arguably the most common inherited tubulopathy, has been linked to a surprising number of genes. The clinical diagnosis rests on a pattern of biochemical abnormalities (hypokalemic, hypochloremic alkalosis with hypomagnesemia, and hypocalciuria) that is specific for impaired salt reabsorption in the distal convoluted tubule (DCT). The same pattern is seen with the clinical use of thiazides, which block the sodium-chloride cotransporter (NCC) expressed in the DCT. Thus it was not surprising when variants in the gene encoding this transporter, *SLC12A3,* were identified as causative. Yet subsequently, a number of other genes (*CLCKNB, HNF1B, KCNJ10, FXYD2, KCNJ16, ATPA1*, *MT-TI,* and *MT-TF*) have been associated with the same biochemical abnormalities.[125] In some form or other, all these genes are involved in salt transport in the DCT, so physiologically this makes sense. However, given that variants in some of these genes follow an autosomal recessive, others an autosomal dominant, and yet others a mitochondrial inheritance pattern, as well as that some are associated with additional clinical manifestations, genetic testing is crucial to properly inform patient and family counseling. Tubulopathies are discussed further in Chapter 71 and in dedicated chapters related to electrolyte transport.

MONOGENIC URINARY STONE DISEASES

The section above “Common and Rare non-Mendelian Contributors to Urinary Stone Disease” considered the contribution of DNA sequence variants association to non-Mendelian heritability of kidney and urinary tract stone diathesis. In the case of monogenic inheritance, the diagnostic yield in urinary stone disease is quite variable and dependent on the underlying problem. In disorders with a highly specific phenotype and only a few associated genes, such as cystinuria, xanthinuria, or adenine phosphoribosyltransferase deficiency, the yield is similarly high as with tubulopathies. The clinical analysis is easily established by analysis of stone composition and/or the highly elevated concentrations of the respective stone-forming substances in the urine. There is also no overlap with idiopathic stones, as these are typically calcium containing.[126] Thus in these disorders, genetic testing has a similar role as in tubulopathies by confirming the diagnosis and informing patient and family counseling. There are also treatments available (e.g., urinary alkalinization and cystine-binding drugs for cystinuria), so the diagnosis also has therapeutic consequences.

Yet as noted earlier, the vast majority of stones, especially those first presenting in adulthood, are not caused by Mendelian disorders but rather a combination of multiple environmental (e.g., diet and climate) and genetic factors, such as variants in *SLC34A3* as discussed earlier. Thus in stone formers without clinical suspicion of an underlying Mendelian disorder, genetic testing is challenging, as the *a priori* probability of identifying an underlying genetic cause is low. For instance, in a cohort from Switzerland of

787 adults with recurrent stone disease, the yield of genetic testing was 2.9% (N = 23), the majority (N = 13; 57%) of whom had cystinuria.[127] Importantly, in 13% (N = 3) a diagnosis of primary hyperoxaluria (PH) was made. PH is a metabolic disorder due to defects in specific enzymes in the liver leading to an overproduction of oxalate. This then needs to be excreted in the urine, with the risk of nephrocalcinosis and stone formation.[128] Currently, three different forms are recognized and associated with distinct genes (*AGXT, GRHPR,* and *HOGA1*), all involved in oxalate metabolism in the liver. Kidney failure from recurrent stone disease can result and in the most severe cases kidney failure from oxalate deposition can occur even in infancy. As the kidney is only secondarily involved (the primary defect is in the liver), a kidney transplant is also at risk of failure from oxalosis. Importantly, for the most common and typically most severe form, PH1, treatment is now available in the form of a small interfering RNA (siRNA), Lumasiran, that inhibits the enzyme upstream of AGXT, thereby suppressing metabolite delivery for oxalate production. Another siRNA, Nedosiran, inhibiting the liver isoenzyme of lactate dehydrogenase (LDH), the final pathway for all forms of PH, is being trialed and could be a treatment for all forms of PH.[129] Liver transplantation is another option in severe cases. Thus making the diagnosis is important because PH can be a severe disease leading to kidney failure, and treatment is available (PH is discussed further in Chapters 40 and 71). Therefore should all stone-forming patients be genetically screened? If the Swiss data are representative, then the a priori likelihood of finding a causative PH variant is 3/787 = 0.4%. And this was a cohort of recurrent stone formers (i.e., selected for more severe disease). If all incident stone formers were assessed, the yield would likely drop even further. Thus a more selective approach appears justified, testing only cases with a high clinical likelihood of having underlying PH, and indeed this has been recommended for stone disease in general by focusing on younger patients and those with a metabolic evaluation (stone composition and urine metabolic analysis) suggestive of an underlying metabolic disorder.[130] Unfortunately, calcium-oxalate stones are common and hyperoxaluria is often secondary, due to diet or increased intestinal absorption of oxalate in the context of intestinal disorders, such as short-gut syndrome or inflammatory bowel disease.[131] A highly specific selection, such as in cystinuria, is therefore difficult and even by focusing on patients with increased urinary oxalate, the diagnostic yield will be low for PH. As population-level sequencing data become available, it is now clear that both rare and common genetic variations can contribute to the risk of kidney stones, with recent identification of monoallelic rare, damaging variants in *SLC34A3* (encoding a sodium-dependent phosphate transporter) present in around 5% of stone formers and 1.6% of controls, giving an odds ratio (and 95% confidence interval) of such variants of 3.75 (2.27–5.91), similar in magnitude to the top decile of polygenic risk for kidney stones, and accounting for almost 10% of the previously missing heritability of urinary stone disease.[40] While such variants (many of which predict truncation of the protein) can be reliably ascertained in an individual, their effect on clinical risk of exhibiting the trait (i.e., kidney stones) is so modest that they are not associated with inherited (i.e., autosomal dominant) kidney stone disease. Thus while "pathogenic" in the sense that they demonstrably contribute to risk, they are not usefully regarded as causative of a Mendelian disease and hence would not constitute a diagnostic finding (although it should be noted that biallelic damaging *SLC34A3* variants cause autosomal recessive hereditary hypophosphatemic rickets with hypercalciuria, which clearly is a Mendelian trait[41]). Kidney stones are discussed further in Chapter 40.

INHERITED KIDNEY TUMOR SYNDROMES INCLUDING TUBEROUS SCLEROSIS

While the kidney is the 8th commonest site of cancer in the United States, affecting around 82,000 people each year,[132] kidney cancer can also rarely occur as the result of an underlying inherited disorder, with a number of different autosomal dominant syndromes described. These include von Hippel-Lindau (VHL) disease, hereditary leiomyomatosis and renal cell carcinoma (HLRCC) syndrome, and Birt Hogg Dube (BHD) syndrome. Each of these disorders has characteristic histologic or extrarenal features, and making a precise diagnosis is critical for offering optimal management. This is firstly because it allows screening for extrarenal features of disease (such as pheochromocytoma in VHL) and also because the precise diagnosis directly informs treatment and prevention strategies for kidney cancers. While the chance of metastasis in clear cell renal cell carcinomas (RCC) that measure <3 cm in diameter is considered to be low (whether occurring as a sporadic lesion or in a patient with VHL disease), in patients with HLRCC, the typical type 2 papillary renal cell carcinomas that they manifest can spread even when the primary lesion is radiologically far smaller. This means that earlier and more surgically aggressive intervention is recommended for patients with radiologic evidence of renal lesions in the setting of this disease. Conversely, in BHD, the lesions tend to be least aggressive on the spectrum (typically chromophobe RCCs that arise from well-differentiated collecting duct cells). The principal renal tumor syndromes, summarized in Table 44.5, are all inherited as autosomal dominant traits, and all are thought to result from a somatic second hit mechanism apart from rare, activating *MET* variants in hereditary papillary renal cell carcinoma syndrome.

The identification of linkage of von Hippel-Lindau disease with the rare variant of the *VHL* gene,[133] the discovery of the role of its protein product (VHL) in targeting for proteolysis hydroxylated HIF α subunits,[13] and the appreciation of the broader role of this oxygen-sensing system in animal physiology provided key insights into the molecular pathogenesis of not only hereditary kidney cancers but also sporadic clear cell renal cell carcinoma (of which approximately 80% show biallelic somatic *VHL* loss) and other cancer types. Indeed, it was revealed that there is a fundamental physiologic process operating in almost all types of multicellular organisms (leading to the award of the 2019 Nobel Prize in Physiology or Medicine to William Kaelin Jr, Gregg Semenza, and Peter Ratcliffe, a nephrologist). These developments provide an exemplar for the paradigm of rare, extreme genetic variation manifesting as a Mendelian trait unveiling biological mechanisms underlying numerically important common diseases. More recently, these advances have been translated into a targeted

Table 44.5 Inherited Kidney Tumor Syndromes

Syndrome	Gene	Mode of Inheritance	Renal and Histologic Features	Extrarenal Features	Notes
von Hippel-Lindau (VHL) disease	*VHL*	AD	Clear cell RCC; renal cysts	CNS hemangioblastoma; retinal angiomas; pancreatic, renal, and epididymal cysts; pheochromocytoma and other neuroendocrine tumors	Systemic treatment with HIF2A inhibitor belzutifan now licensed
Hereditary leiomyomatosis and RCC (HLRCC) syndrome	*FH*	AD	Type 2 papillary RCC, renal cysts	Uterine fibroids; painful cutaneous leiomyomas	Biallelic FH mutations associated with severe metabolic disease fumarate hydratase deficiency
Birt Hogg Dube (BHD) syndrome	*FLCN*	AD	Chromophobe or oncocytoma, renal cysts	Pulmonary cysts and pneumothorax; painless cutaneous fibrofolliculomas, colonic polyps	RCC occurs in <20% of those affected
Hereditary papillary renal cell carcinoma (RCCP1) syndrome	*MET*	AD	Multifocal type 1 papillary RCC	None described	Highly penetrant activating mutations
Phaeochromocytoma/ Paraganglioma (PPS) syndrome	*SDHB* *SDHC* *SDHD,* *SDHA*	AD	SDHB-deficient RCC (on immunostaining)	Phaeochromocytoma; Paraganglioma	*SDHA* is least common; biallelic *SDHA* mutations associated with Leigh syndrome
Tuberous sclerosis complex	*TSC1* *TSC2*	AD	Angiomyolipoma, renal cysts (rarely oncocytoma or aggressive epithelioid AML)	Facial angiofibromas, ungual fibromas, shagreen patch, hypomelanotic macules; seizures, developmental delay, cortical tubers, subependymal giant cell astrocytomas; pulmonary lymphangioleiomyomatosis (mostly in females); cardiac rhabdomyomas in infancy; retinal hamartomas in childhood	AMLs are primary cause of mortality in adults with TSC. Suspect malignant epithelioid AML if rapidly enlarging AML. Systemic treatment with mTOR inhibitor Everolimus now licensed
Cowden syndrome	*PTEN*	AD	Multifocal RCC	Macrocephaly; GI polyps; breast, thyroid, endometrial tumors	

AD, Autosomal dominant; *AML,* angiomyolipoma; *CNS,* central nervous system; *RCC,* renal cell carcinoma.

treatment, with approval in the United States of belzutifan (a HIF 2α inhibitor) for treatment of advanced (sporadic) kidney cancer,[134] as well as for treatment of kidney (and other) tumors in people with VHL disease.[135] Since until now, treatment of kidney cancers in VHL disease has principally been with nephron-sparing surgery (associated with a high risk of both kidney failure and metastatic disease), the availability of targeted medical therapy could result in improved outcomes for people with this disease.

Tuberous sclerosis complex (TSC) frequently presents in infancy or early childhood with neurologic or cardiologic manifestations. However, during childhood, adolescence, or adulthood approximately 80% of patients will eventually develop (usually nonmalignant) renal lesions called angiomyolipomas (AMLs). While they are small, these lesions are unlikely to cause harm, but when they grow to more than 4 cm in diameter, and especially if there are enlarged or pseudoaneurysmal blood vessels, the risk of spontaneous hemorrhage develops. Since hemorrhage from an enlarged AML is potentially fatal (and in fact historically, renal involvement was the leading cause of death among people with TSC[136]), surveillance and consideration of prophylactic treatment of AMLs >4 cm is recommended. Appreciation of the role of hamartin and tuberin (the protein products of the *TSC1* and *TSC2* genes, respectively) in breaking down mammalian target of rapamycin (mTOR)[137] led to clinical trials of the mTOR inhibitor everolimus that justified approval of the use of this drug to treat renal AMLs[138] (and more recently epilepsy[139]) in TSC, broadening the preventive treatment modalities beyond invasive surgical or radiologic embolization therapies previously available that often reduce nephron number.

UNEXPLAINED KIDNEY FAILURE

Data from population-based registries indicate that around 20% of adults receiving renal replacement therapy do not

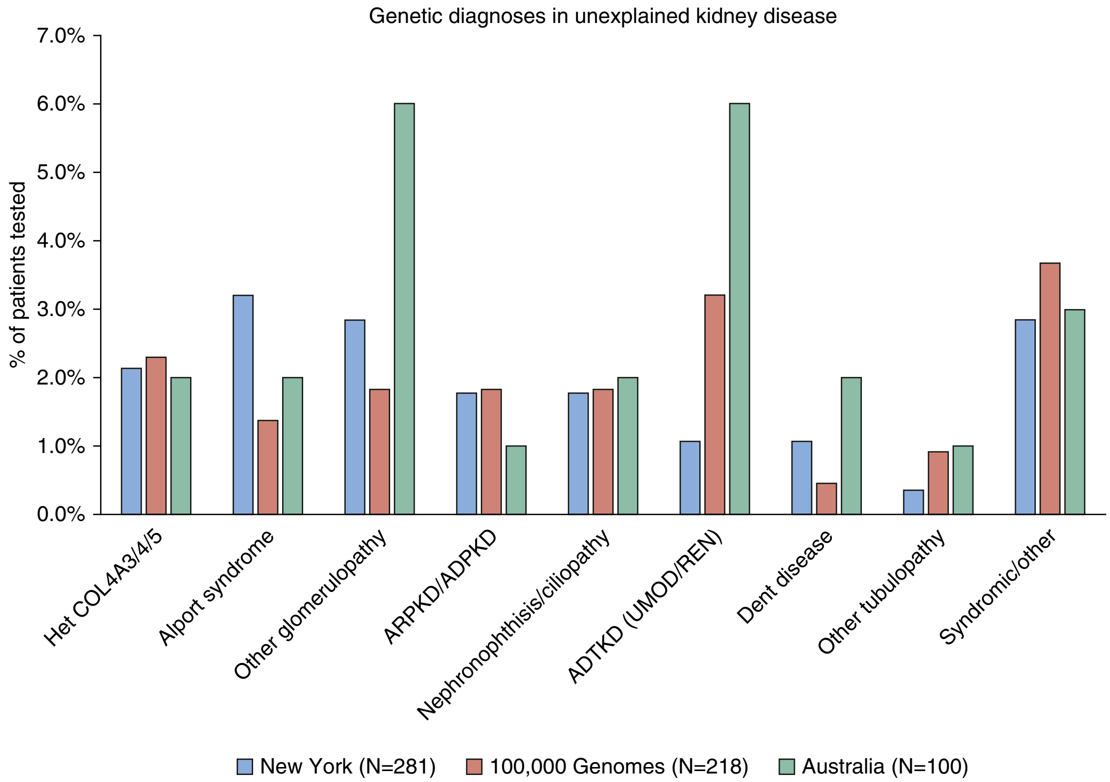

Fig. 44.6 Genetic diagnoses in patients undergoing genomic testing for unexplained kidney failure or nephropathy of unknown cause in New York,[3] the UK 100,000 Genomes Project, and Australia.[164]

have a diagnosis assigned (variously termed unexplained kidney failure or nephropathy of unknown cause).[140] This is often because clinical presentation is with advanced renal impairment and scarred kidneys (typically manifesting as small, echo-bright kidneys on an ultrasound scan), at which time kidney biopsy is hazardous and, where performed, may not reveal an underlying diagnosis. Genomic testing in cohorts of such patients shows that between 15% and 25% will have a detectable monogenic disorder.[3,141] The proportion with a diagnostic finding is highest among those with kidney failure between the ages of 18 and 35, and the majority of diagnoses are of autosomal dominant or X-linked recessive conditions, with only a small proportion of autosomal recessive. Accordingly, family history is predictive of the presence of a Mendelian disorder (odds ratio 3.3 in one series[141]). The genes reported in patients with unexplained kidney failure or nephropathy of unknown cause encompass almost all types of monogenic kidney disorders, including syndromic or otherwise distinctive conditions (Fig. 44.6).

These data indicate that in some settings, within this patient group, genomic testing can frequently identify an actionable molecular diagnosis and, as a consequence, genomic testing for patients presenting with unexplained kidney failure is becoming incorporated into clinical practice in some health systems (such as the UK's National Health Service Genomic Medicine Service test directory, https://www.england.nhs.uk/publication/national-genomic-test-directories/). As with all genetic variants identified in a patient, careful consideration must be given to use of these genetic findings as a predictive test in relatives who should, as far as possible, be given quantitative information on the risk of kidney disease should they be found to harbor a variant discovered in a relative with kidney disease. For example, how the finding of an *APOL1* risk genotype should be communicated to a patient with unexplained kidney failure (or their relative) remains unclear. It is also now increasingly recognized that in certain geographic regions, unexplained kidney disease is endemic and clearly the diagnostic value of looking for established Mendelian disease in this setting is likely to be very different.

CONCLUSION

The significant advances in technology that have now made genome-wide genetic variation and genomic sequence data available for analysis in individuals have significantly broadened understanding of how genetic variation causes and contributes to disease, in some cases leading to the development of new, highly effective treatments. This has allowed, in a short space of time, deployment of these technologies in the clinical setting, providing clinicians with powerful tools to better understand kidney disease—both in cohorts and with respect to individual patients. How and when these tools are best used in clinical practice is not completely clear in all diseases, but it seems likely that as more quantitative data linking genetic information with disease risk become available, there will be ever more opportunities for genomic testing to inform clinical decision-making.

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